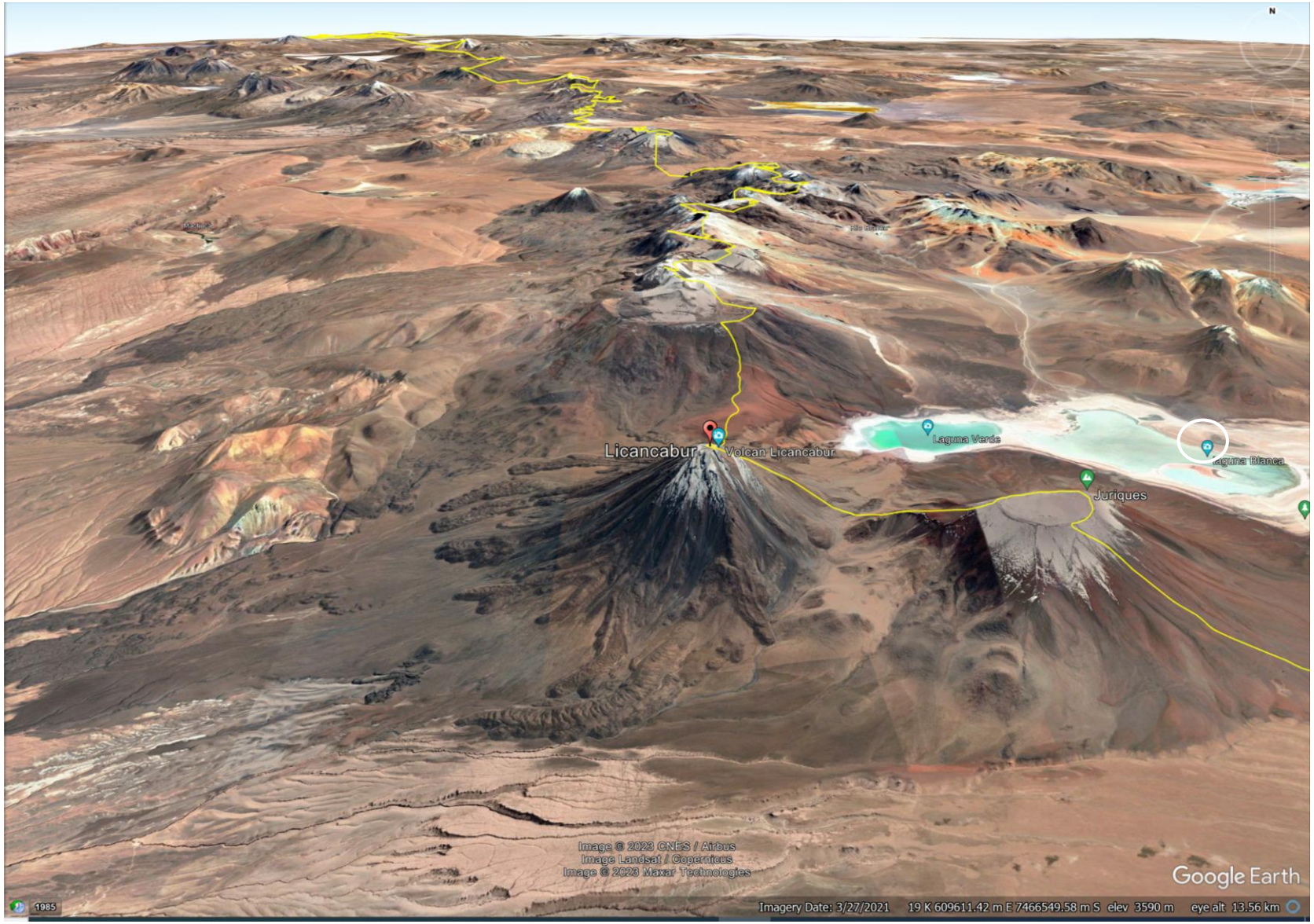
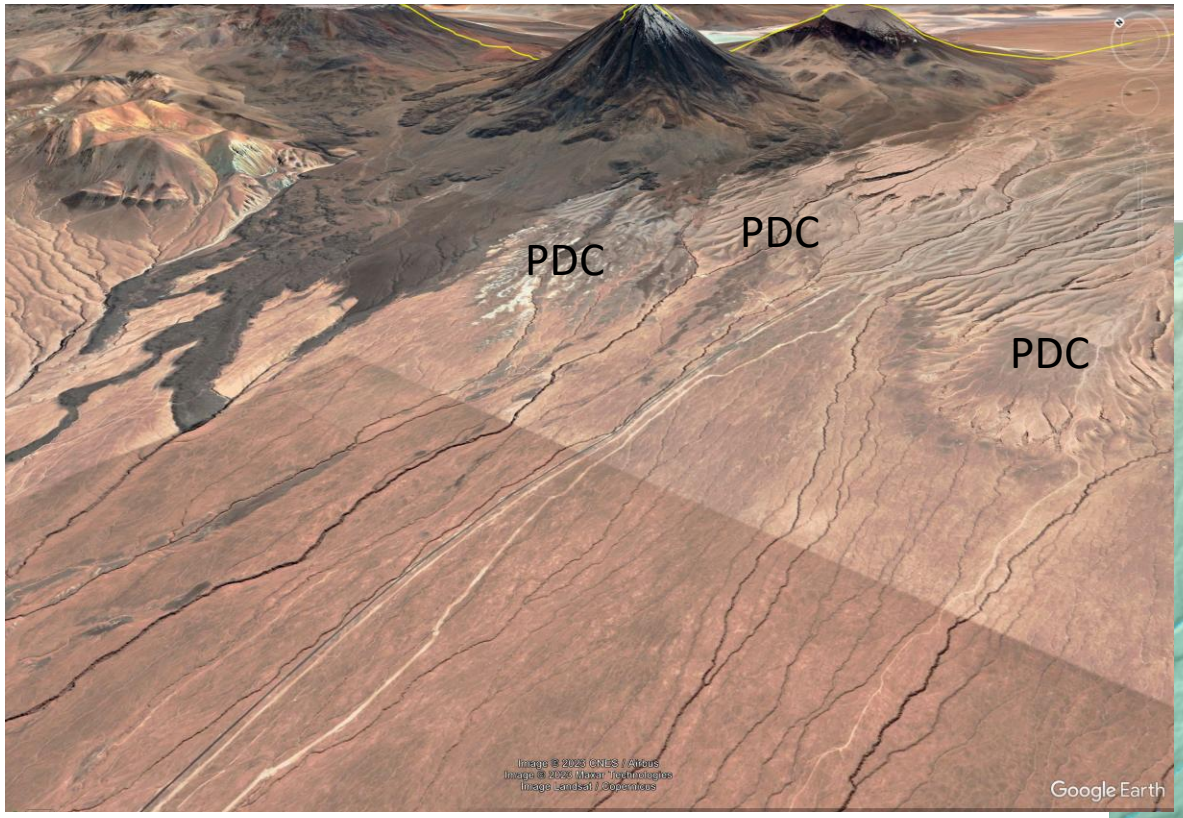


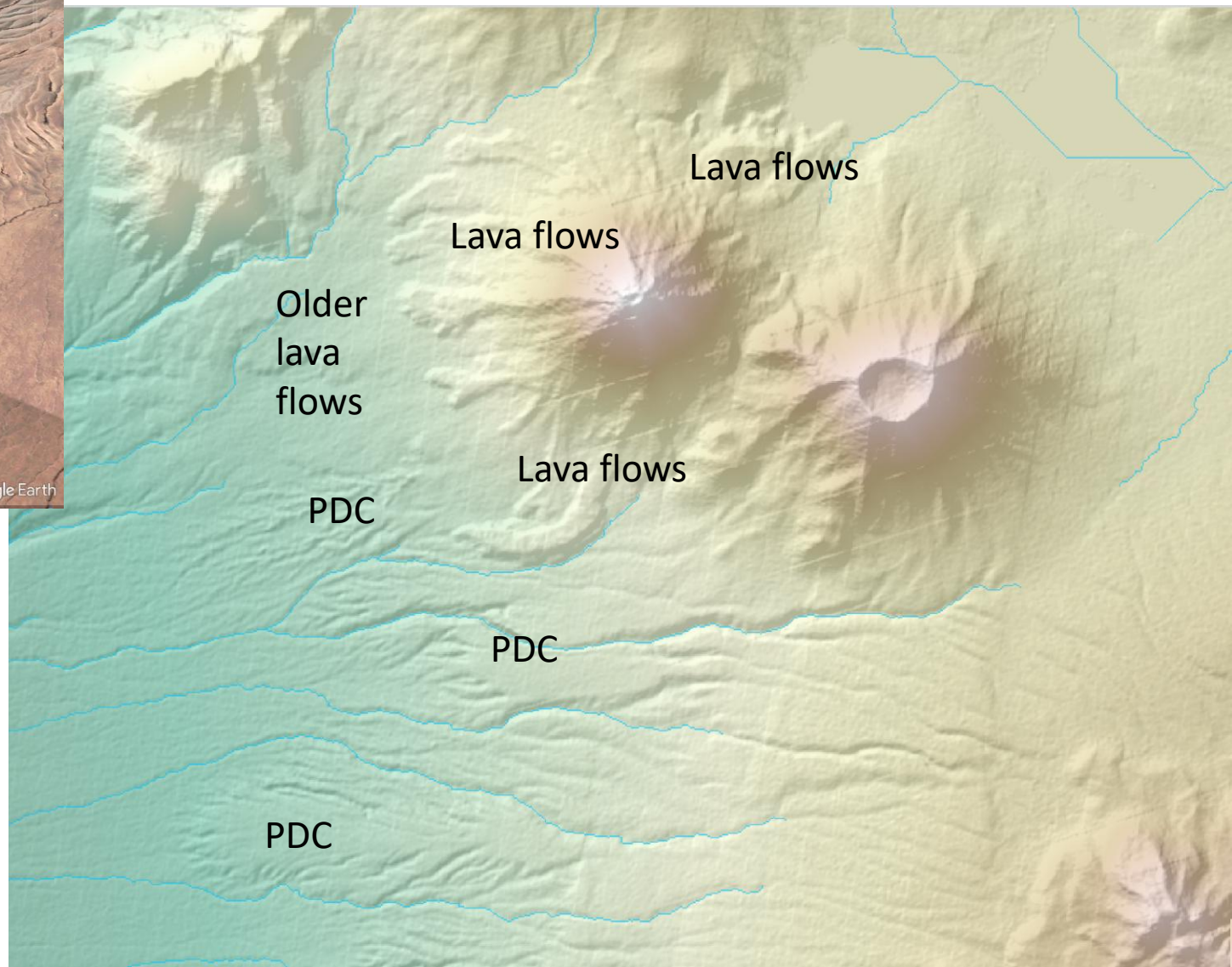
LICANCABUR Energy Cone Exercise



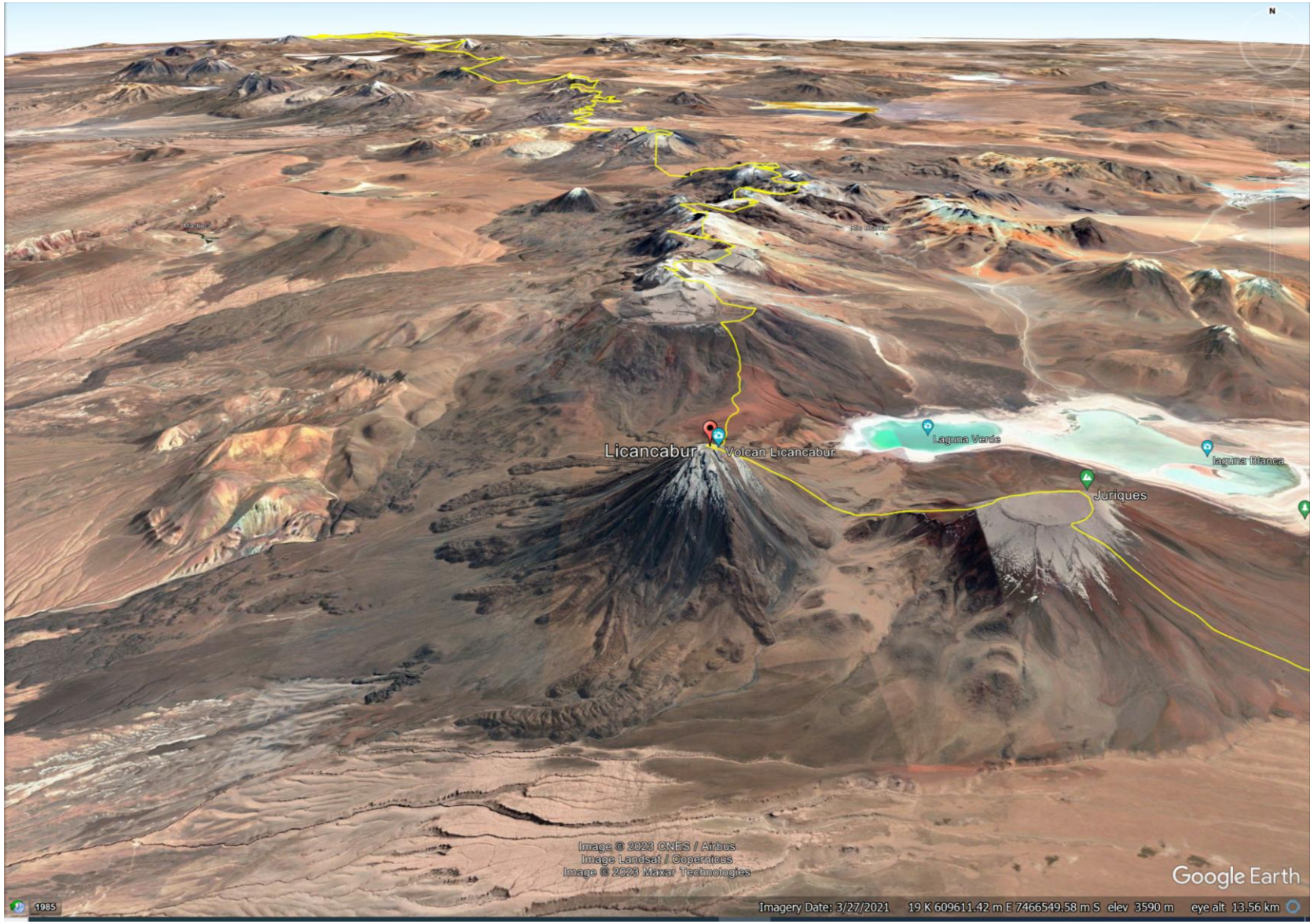


Google Earth and satellite imagery

Geomorphology seen in DEM and shaded-relief map



PDCs = Pyroclastic density currents



These exercises just use Licancabur as an example!

The model parameters in all the following exercises may or may not be realistic.

Our exercise hazard maps are certainly worse than any made by Chileans!!!!!

Licancabur



Country	Chile - Bolivia
Volcanic Region	Central Andean Volcanic Arc
Landform | Volc Type	Composite | Stratovolcano
Last Known Eruption	Unknown - Evidence Credible
Coordinates	22.83°S, 67.88°W
Elevation	5,916 m (19,409 ft)
Volcano Number	355092

Google Earth Placemark with Features

Cite Volcano Profile

Latest Activity Reports | Weekly Reports | Bulletin Reports | General Information

Eruptive History | Subfeatures | Photo Gallery | Sample Collection | External Sites

General Information

Basic Data

Volcano Number	355092
Last Known Eruption	Unknown - Evidence Credible
Elevation	5,916 m / 19,409 ft
Latitude Longitude	22.83°S 67.88°W

Morphology

Volcano Landform
Composite

Geological Summary

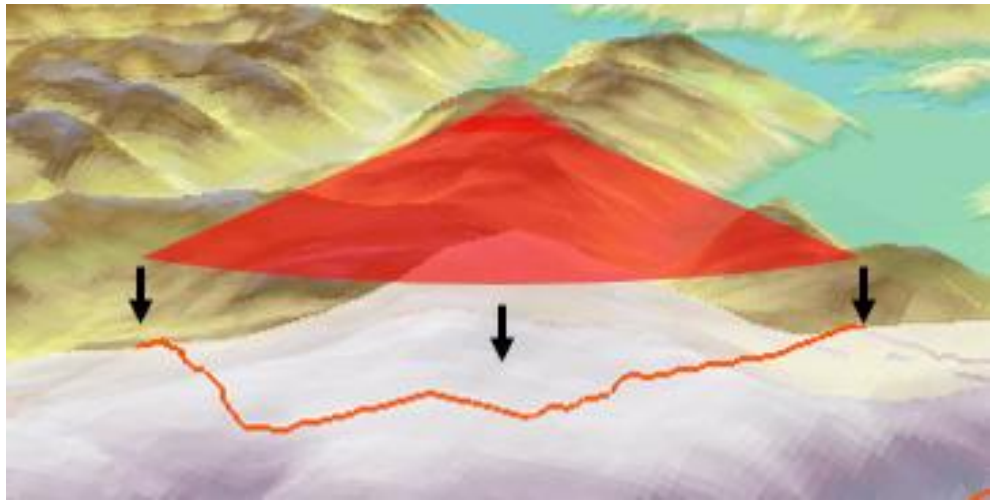
The symmetrical steep-sided Licancabur stratovolcano was constructed primarily during the Holocene and contains a lake in its 400-m-wide summit crater. The Pleistocene Juriques volcano is located immediately to the SE and is capped by a 1.5-km-wide summit crater. Archaeological ruins were found on the crater rim. The shallow freshwater summit lake is 70 x 90 m and has a measured temperature of 6°C, supporting growth of planktonic fauna at nearly 6,000 m elevation. Young blocky andesitic lava flows with prominent levees extend up to 6 km down the NW-to-SW flanks; older flows reach up to 15 km from the summit crater and are covered by pyroclastic-flow deposits that extend 12 km. The most recent activity produced flank lava flows.

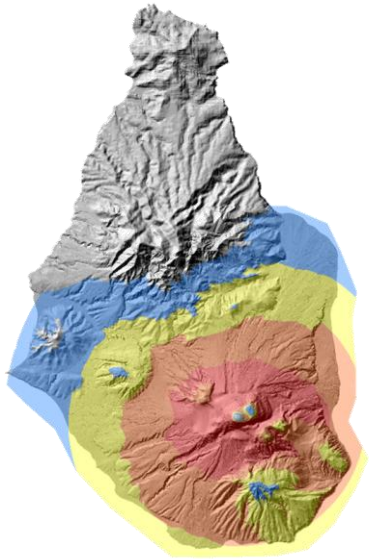
References

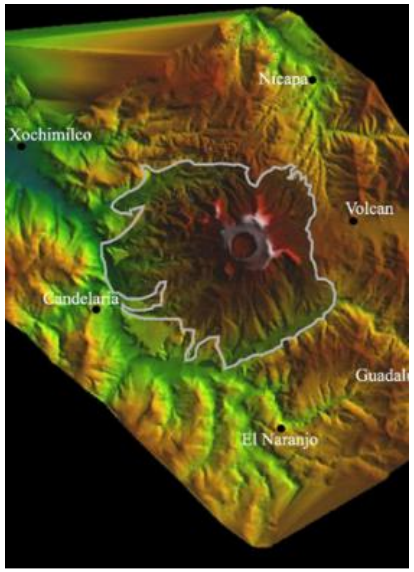
- We are going to use LaharZ to make some more energy cones, but this time, we will use ΔH/L values from FlowDat

Energy Cone

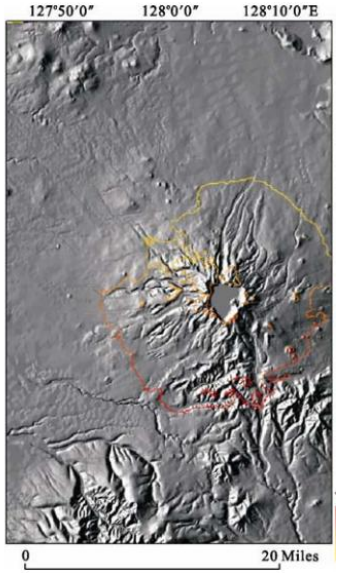
- Also called ΔH/L cone
- One of the first modeling techniques applied to PDCs
- Uses the ΔH/L vs. volume relationship to draw a line from the summit of the volcano to the estimated extent of PDCs using the ΔH/L ratio
- Can be done by hand on a map, or using a GIS program
- Used many places, often in combination with other modeling techniques



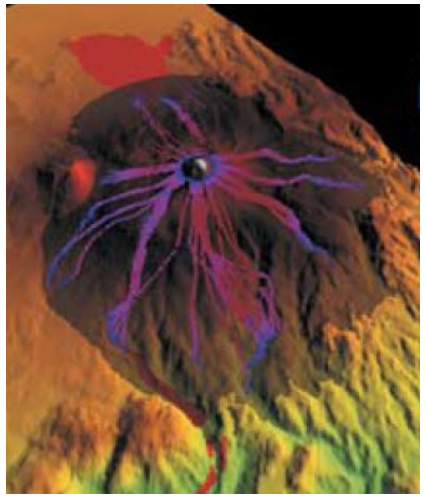




Energy cone at El Chichón, Mexico
Macías et al. 2008



Energy cone at Tianchi volcano, China
Wan et al. 2012



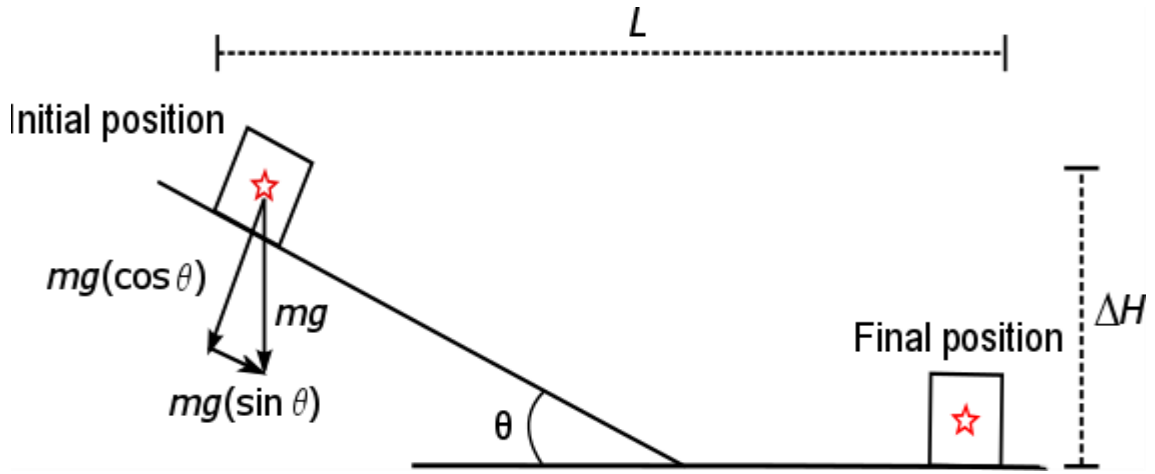
Energy cone & FLOW2D at Citlaltépetl, Mexico
Sheridan & Macias, 2004

VDAP

USGS

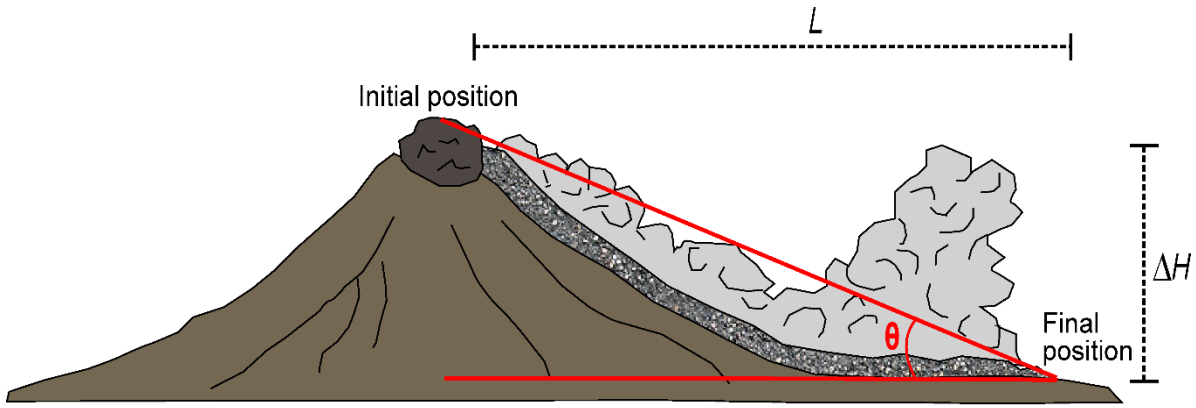
PDC mobility: ΔH/L or Heim coefficient

- Height dropped/runout length
- Block begins to slide when gravitational force down a plane equals the frictional force
- Tanθ =coeff. of friction, μ = ΔH/L = Heim coefficient



Block begins to slide when mgsinθ = μmgcosθ

Coeff. of friction, μ = $\frac{mgsin\theta}{mgcos\theta}$ = tanθ



ΔH/L ranges for dome- and column-collapse PDC energy cones from FlowDat regressions

Note:

- Decimal values are used for energy cones
- Degrees are used for TITAN2D basal friction angles
- The regressions at the bottom are for decimal values
- Use the formula Degrees(atan(ΔH/L value)) to convert decimals to degrees in excel

Volume (km^3)	Volume (10^6 x m^3)	ΔH/L best fit Dome collapse	ΔH/L best fit Dome collapse (degrees)	ΔH/L best fit Column collapse	ΔH/L best fit Column collapse (degrees)
0.0001	0.1	0.39	21.1	0.36	19.8
0.001	1	0.29	16.1	0.30	16.5
0.01	10	0.22	12.1	0.24	13.7
0.1	100	0.16	9.1	0.20	11.4

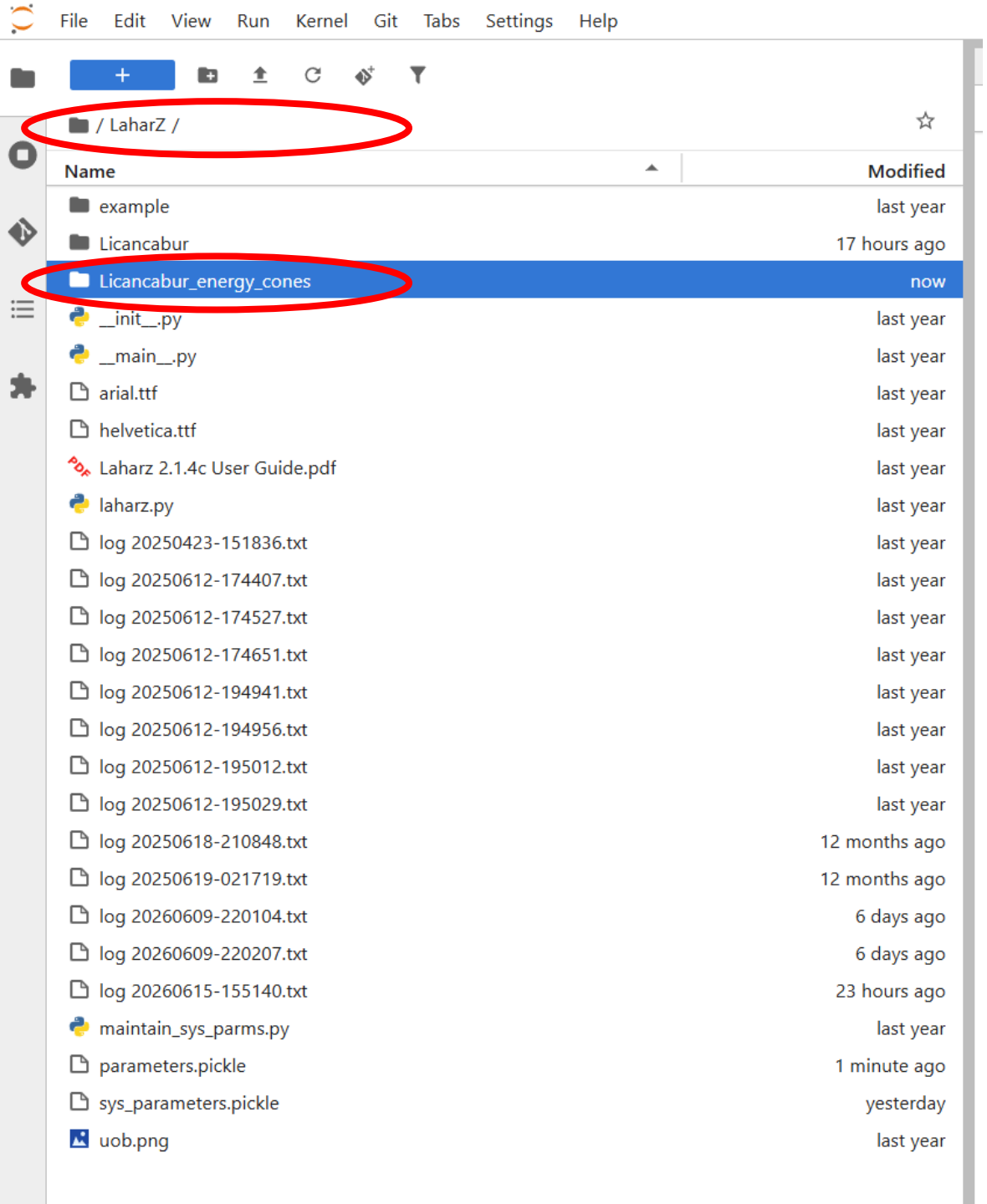
Dome-collapse PDCs (based on 121 flows): $\Delta H/L = 0.120\left(V^{-0.127}\right)$

Column-collapse PDCs (based on 40 flows): $\Delta H/L = 0.166\left(V^{-0.084}\right)$

All PDCs (based on 188 flows): $\Delta H/L = 0.165\left(V^{-0.085}\right)$

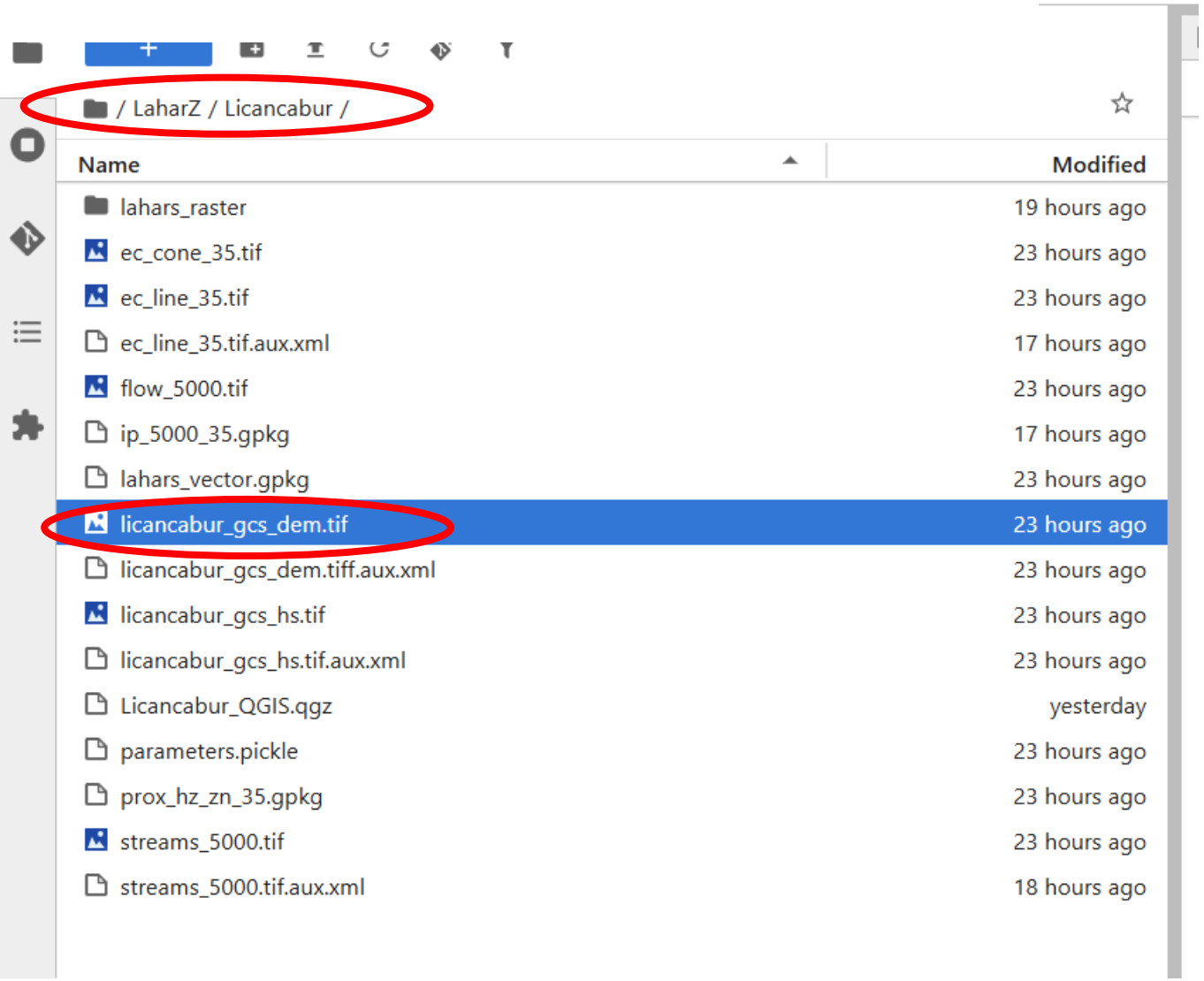
VDAP

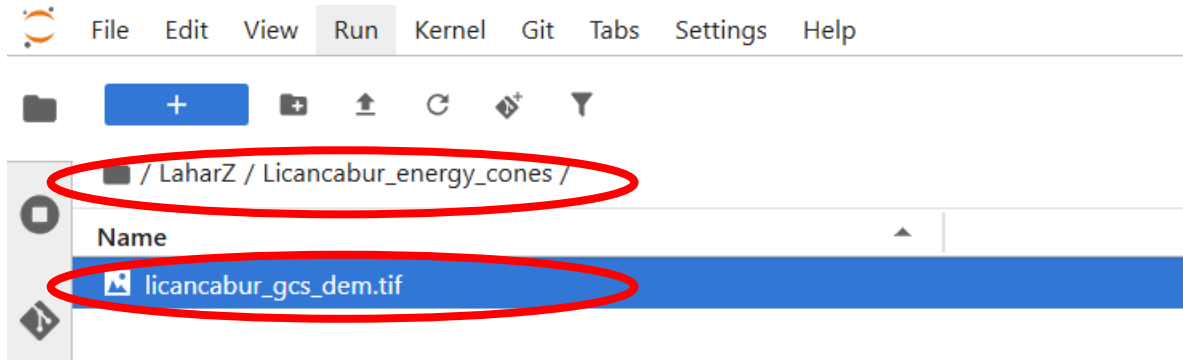
USGS



- We are going to use LaharZ to make some more energy cones, but this time, we will use ΔH/L values from FlowDat
- First, go back to your LaharZ folder
- Make a new folder inside called “Licancabur_energy_cones”

- Copy the DEM you used for modeling lahars into this new folder





ΔH/L ranges for dome- and column-collapse PDC energy cones from FlowDat regressions

- Now let's decide on some volumes and ΔH/L values from FlowDat
- We'll keep things simple and model dome-collapse flows with volumes straight from the table: 0.1, 1, and 10 M m^3
- From the FlowDat regression, we see that these correspond to 0.39, 0.29, and 0.22 ΔH/L values

Volume (km^3)	Volume (10^6 x m^3)	ΔH/L best fit Dome collapse	ΔH/L best fit Dome collapse (degrees)	ΔH/L best fit Column collapse	ΔH/L best fit Column collapse (degrees)
0.0001	0.1	0.39	21.1	0.36	19.8
0.001	1	0.29	16.1	0.30	16.5
0.01	10	0.22	12.1	0.24	13.7
0.1	100	0.16	9.1	0.20	11.4

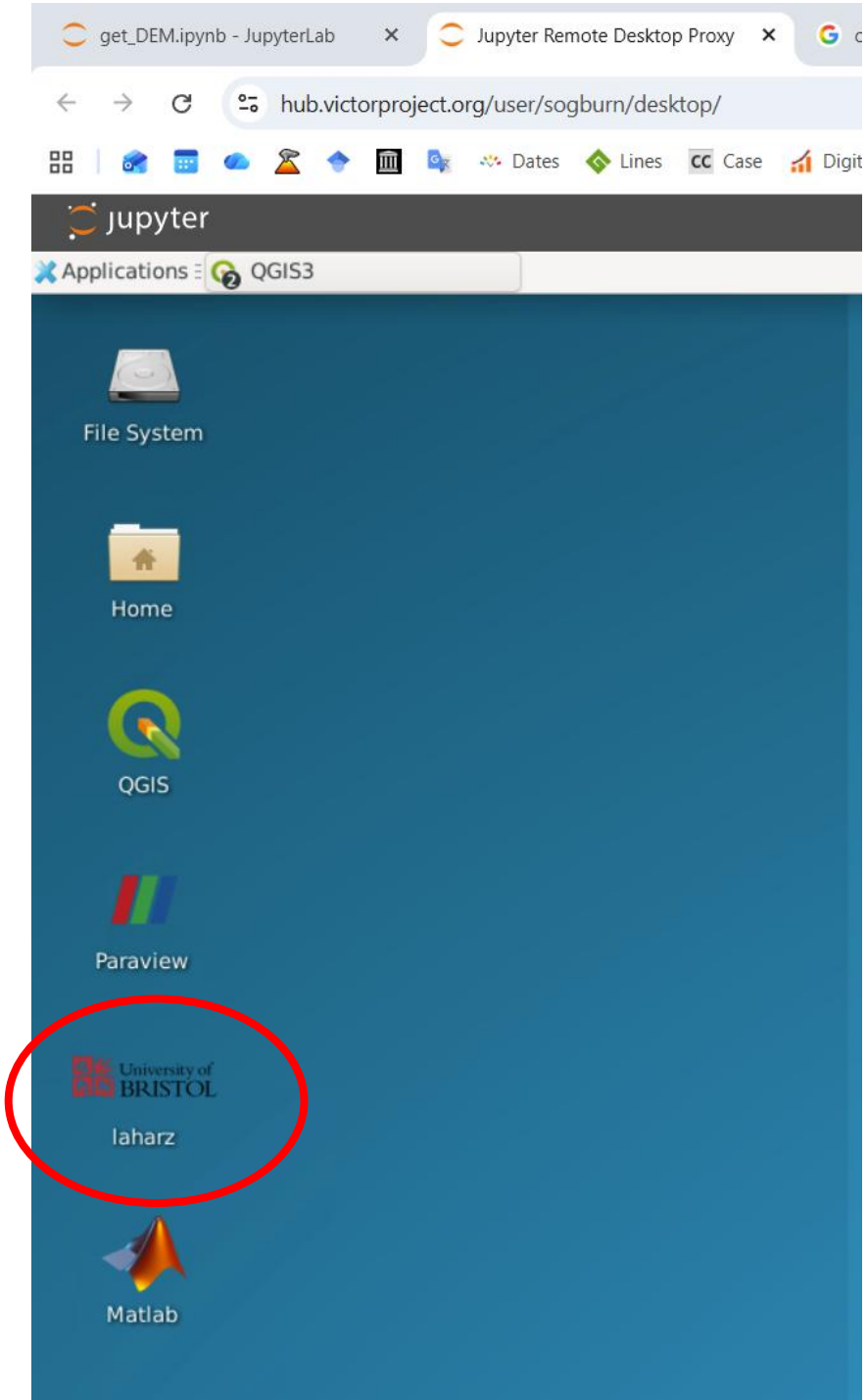
Dome-collapse PDCs (based on 121 flows): $\Delta H/L = 0.120\left(V^{-0.127}\right)$

Column-collapse PDCs (based on 40 flows): $\Delta H/L = 0.166\left(V^{-0.084}\right)$

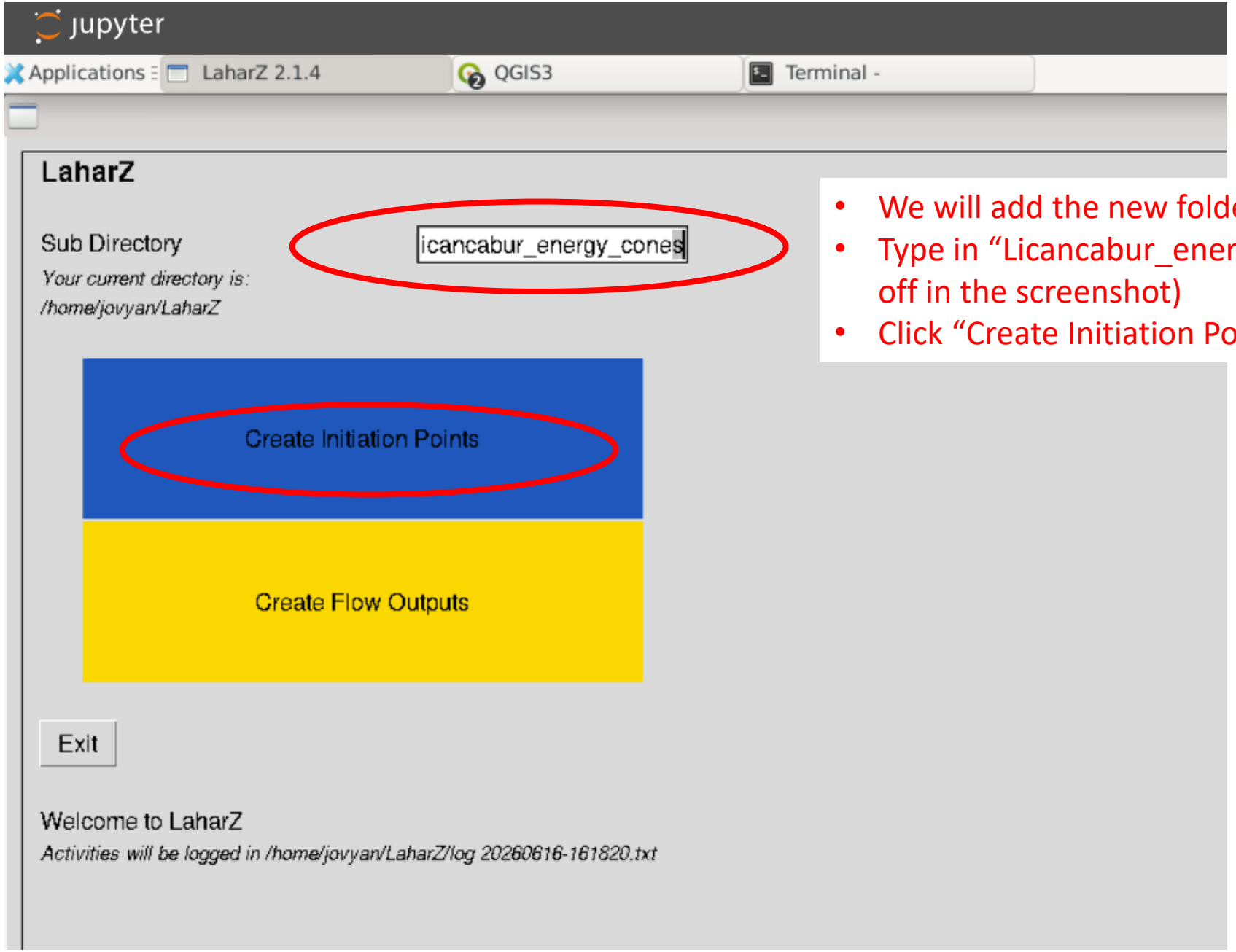
All PDCs (based on 188 flows): $\Delta H/L = 0.165\left(V^{-0.085}\right)$

VDAP

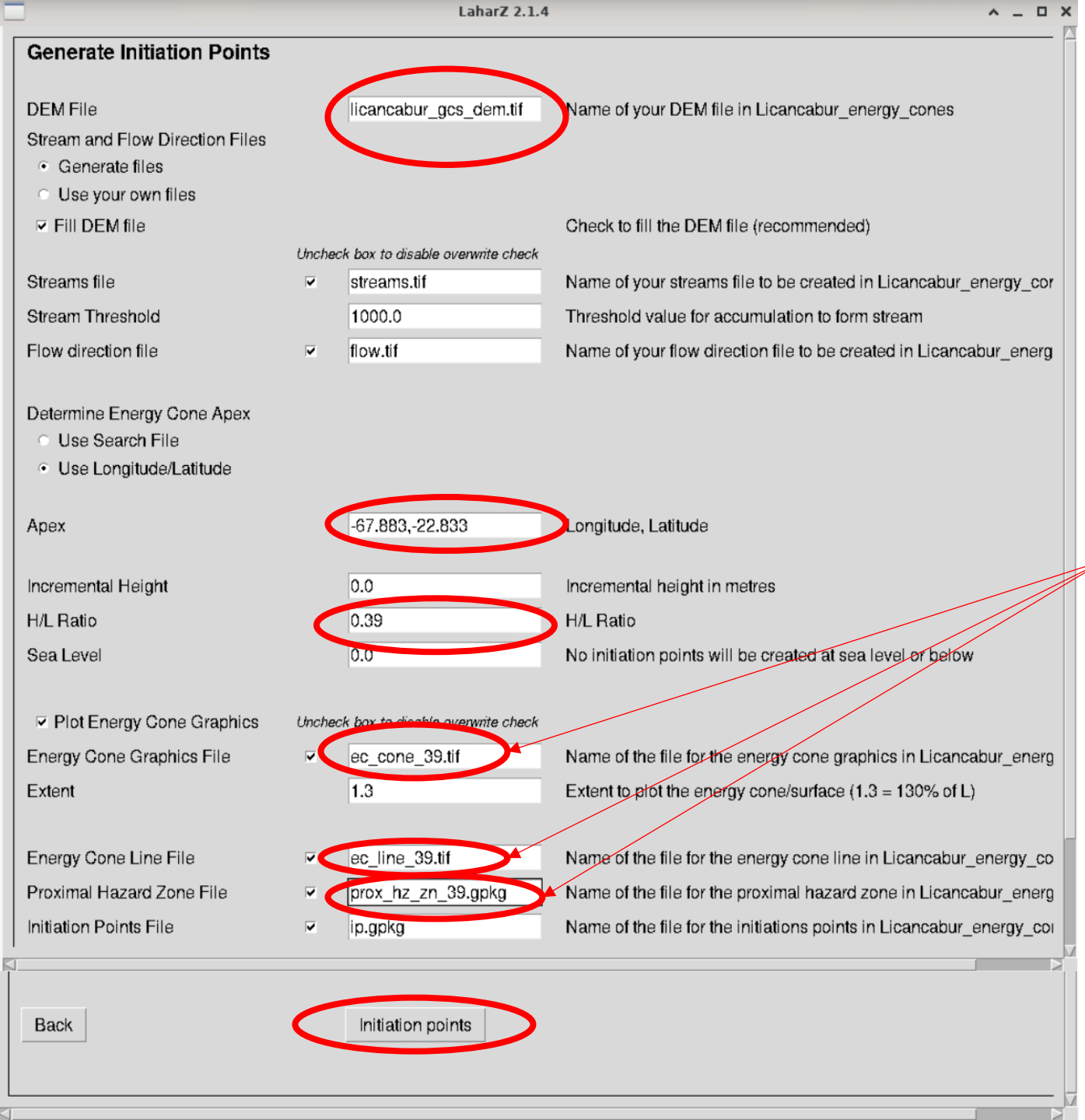
USGS



- Go back to your desktop and open laharz again



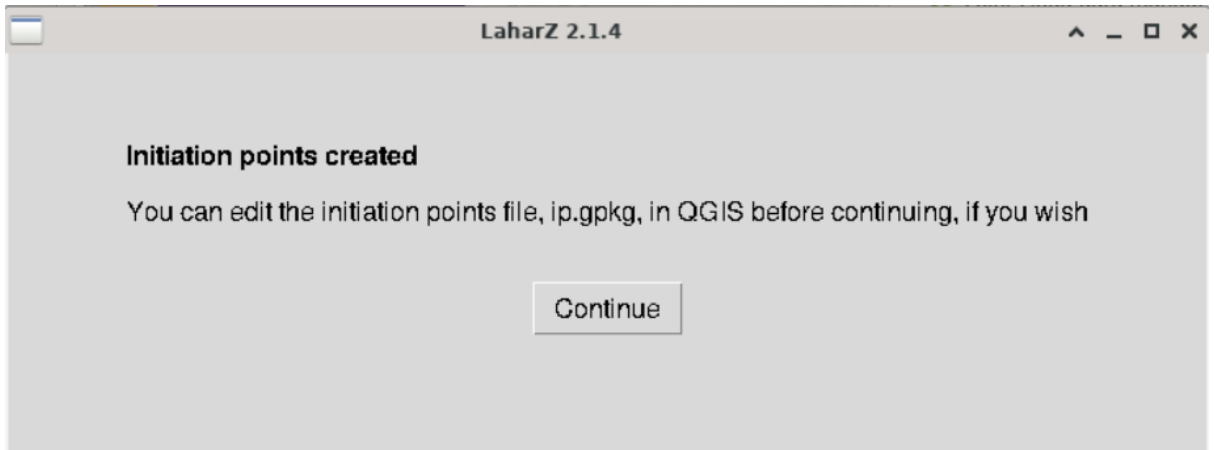
- We will add the new folder we made as the directory
- Type in “Licancabur_energy_cones” as the folder (It’s cut off in the screenshot)
- Click “Create Initiation Points”



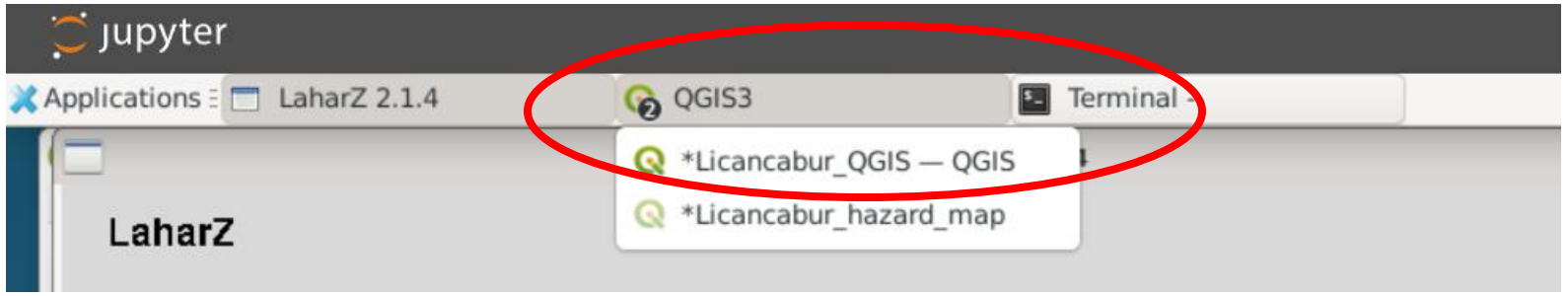
- We used this same screen for LaharZ to make the stream network, the LaharZ maximum hazard zone, and the initiation points
- For modeling PDCs, we don't care about the streams or the initiation points, so you can leave all those as default values

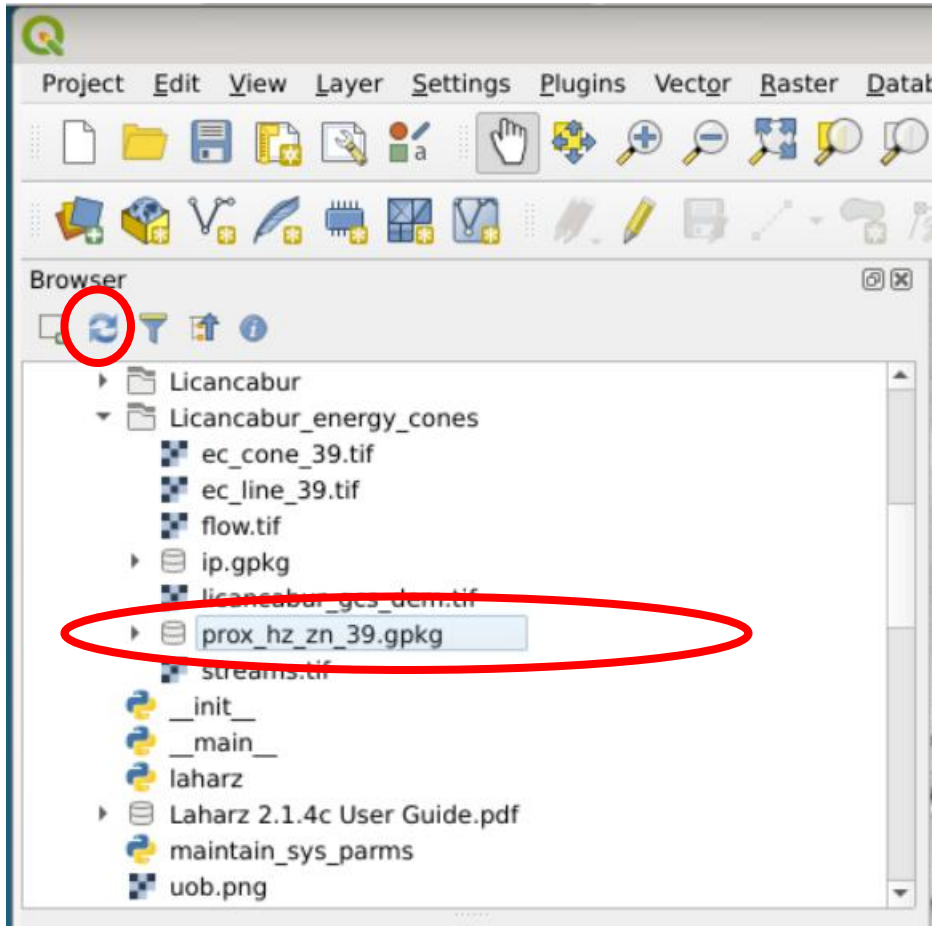
- Change the DEM File to "licancabur_gcs_dem.tif
- Add the Long/Lat coordinates: -67.883, -22.833
- Change the ΔH/L ratio to the FlowDat value for a 0.1 Mm^3 flow (0.39)
- Change all the energy cone files to include "39"

- Click "Initiation points" at the bottom

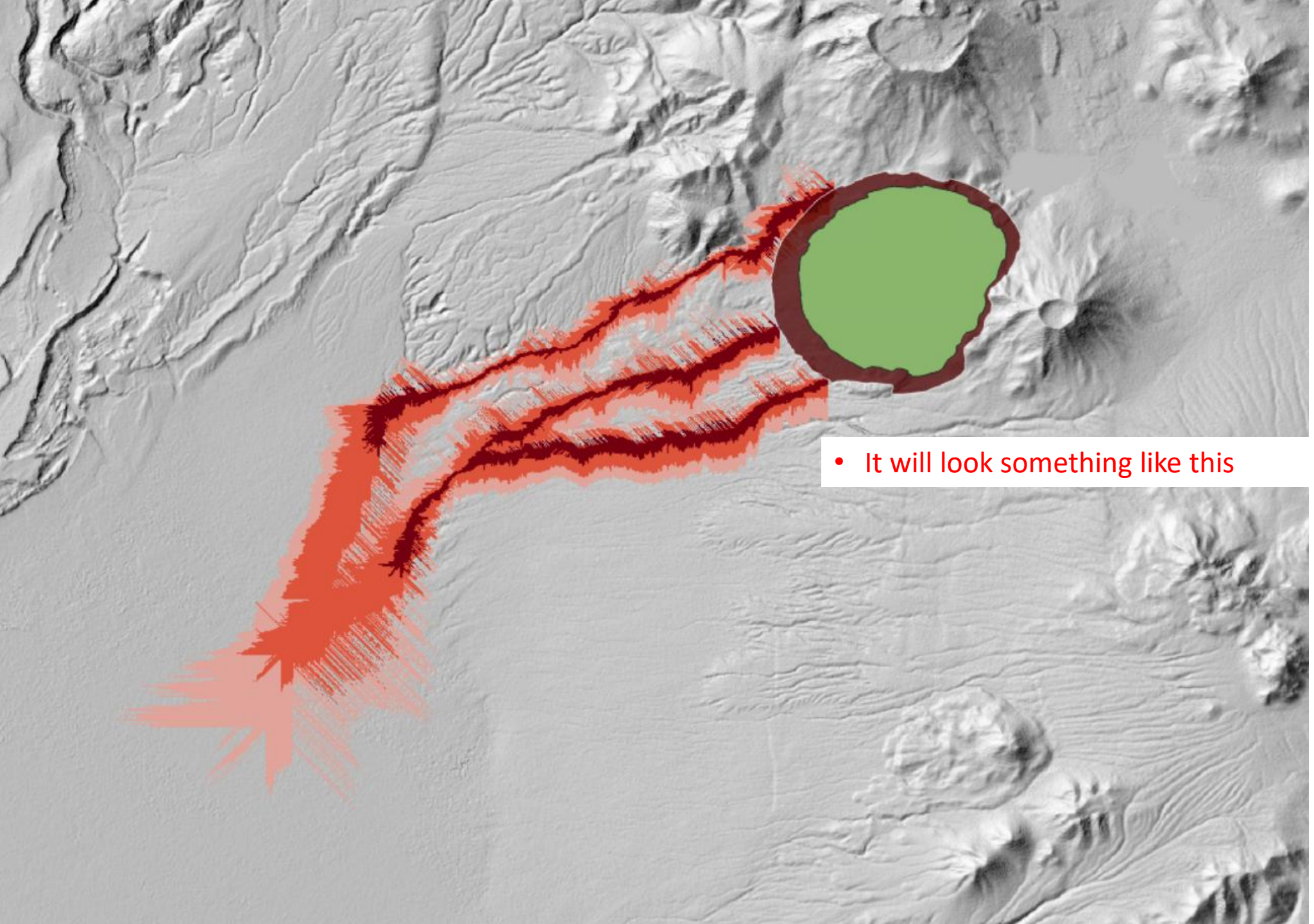


When it’s done, click continue

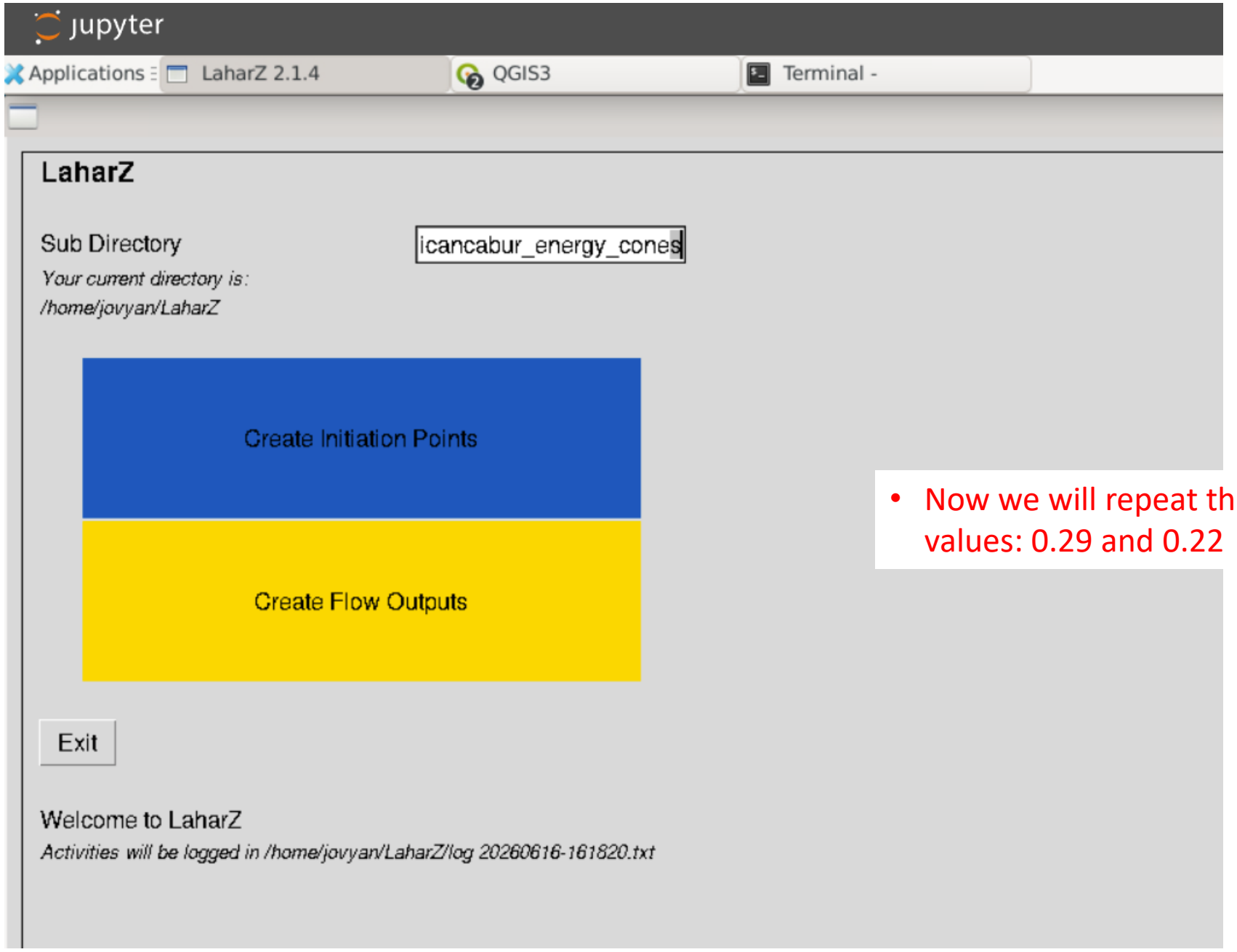




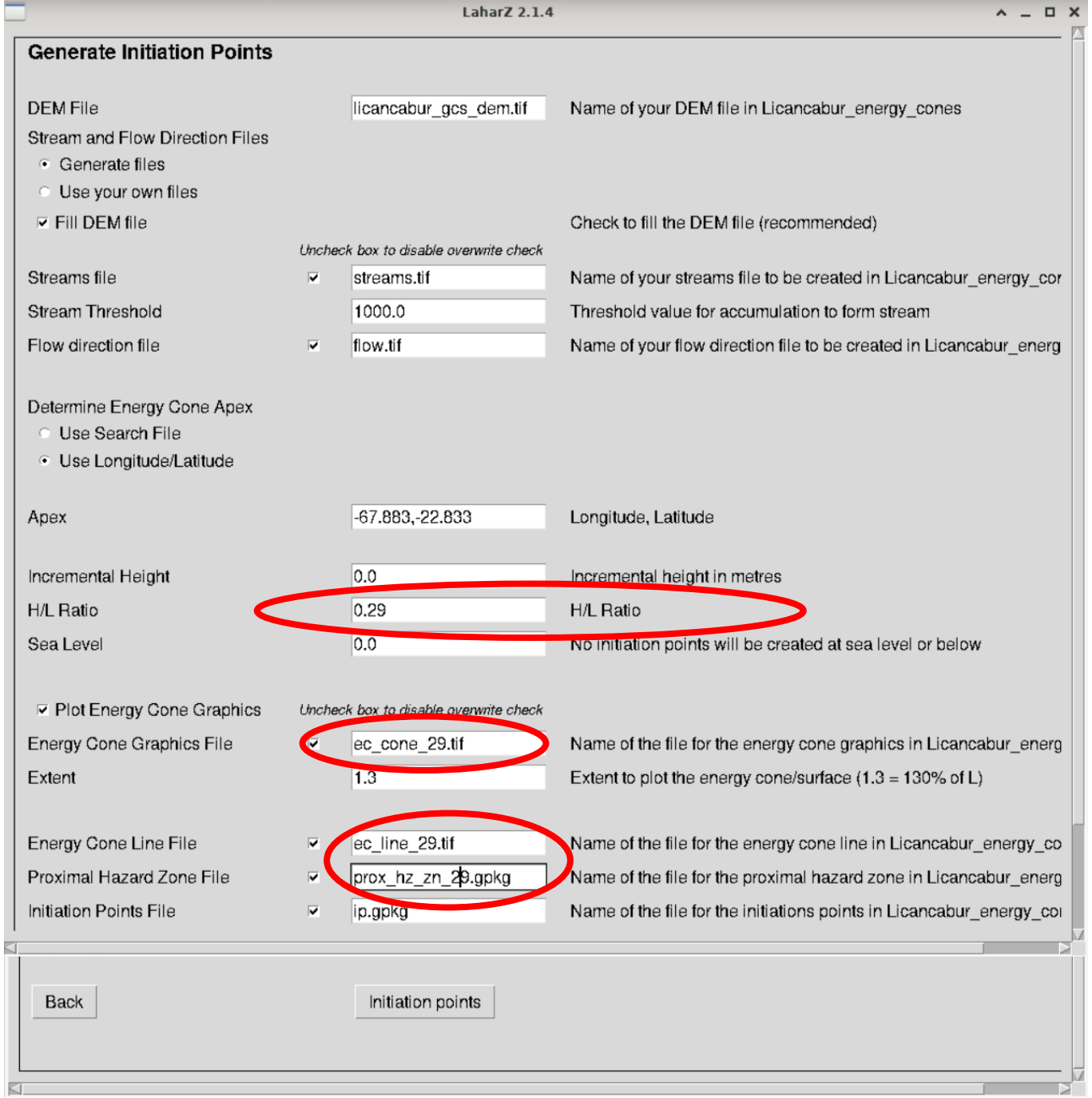
- Let's go look at the results!
- Navigate to QGIS
- Refresh the files if needed
- Add the "prox_hx_zn_39.gpkg" file to the map



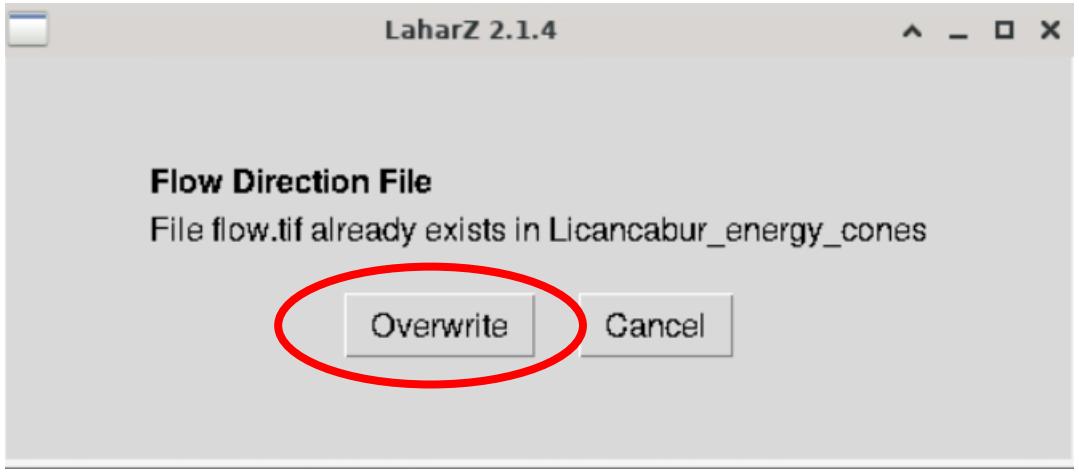
- It will look something like this



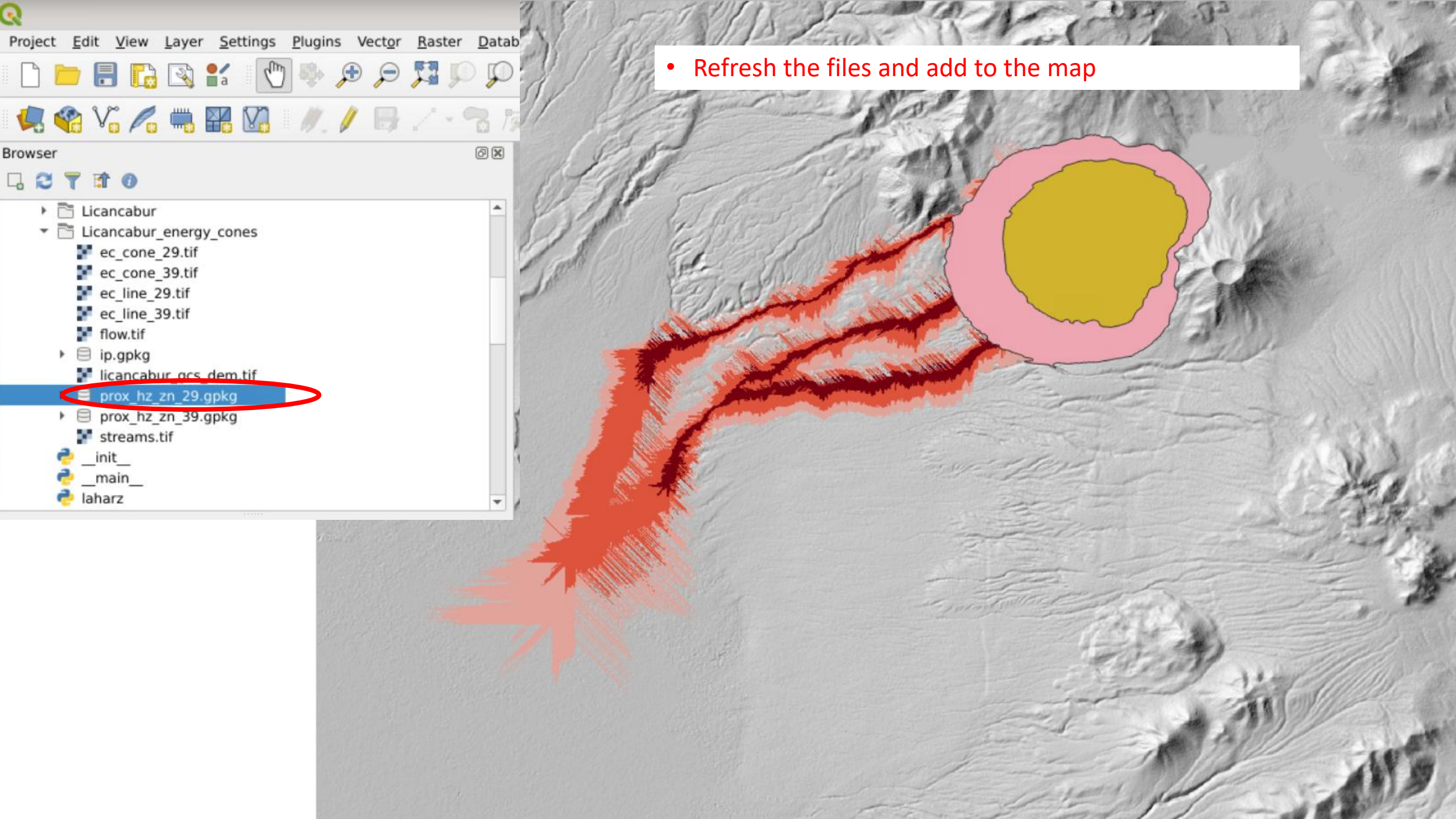
- Now we will repeat this process for the other ΔH/L values: 0.29 and 0.22

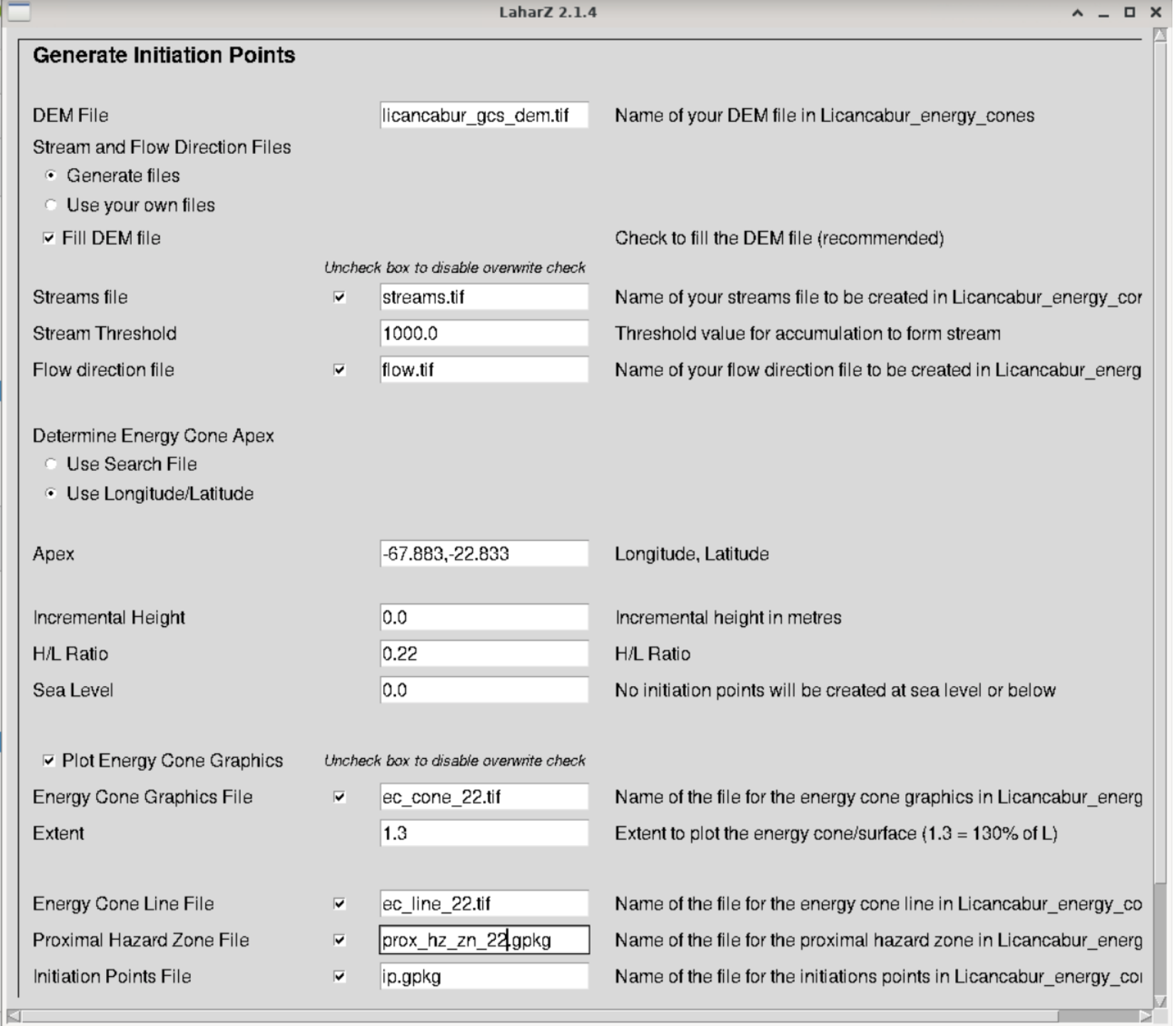


- You can leave everything the same, except change all the ΔH/L related values to 0.29
- Click Initiation Points

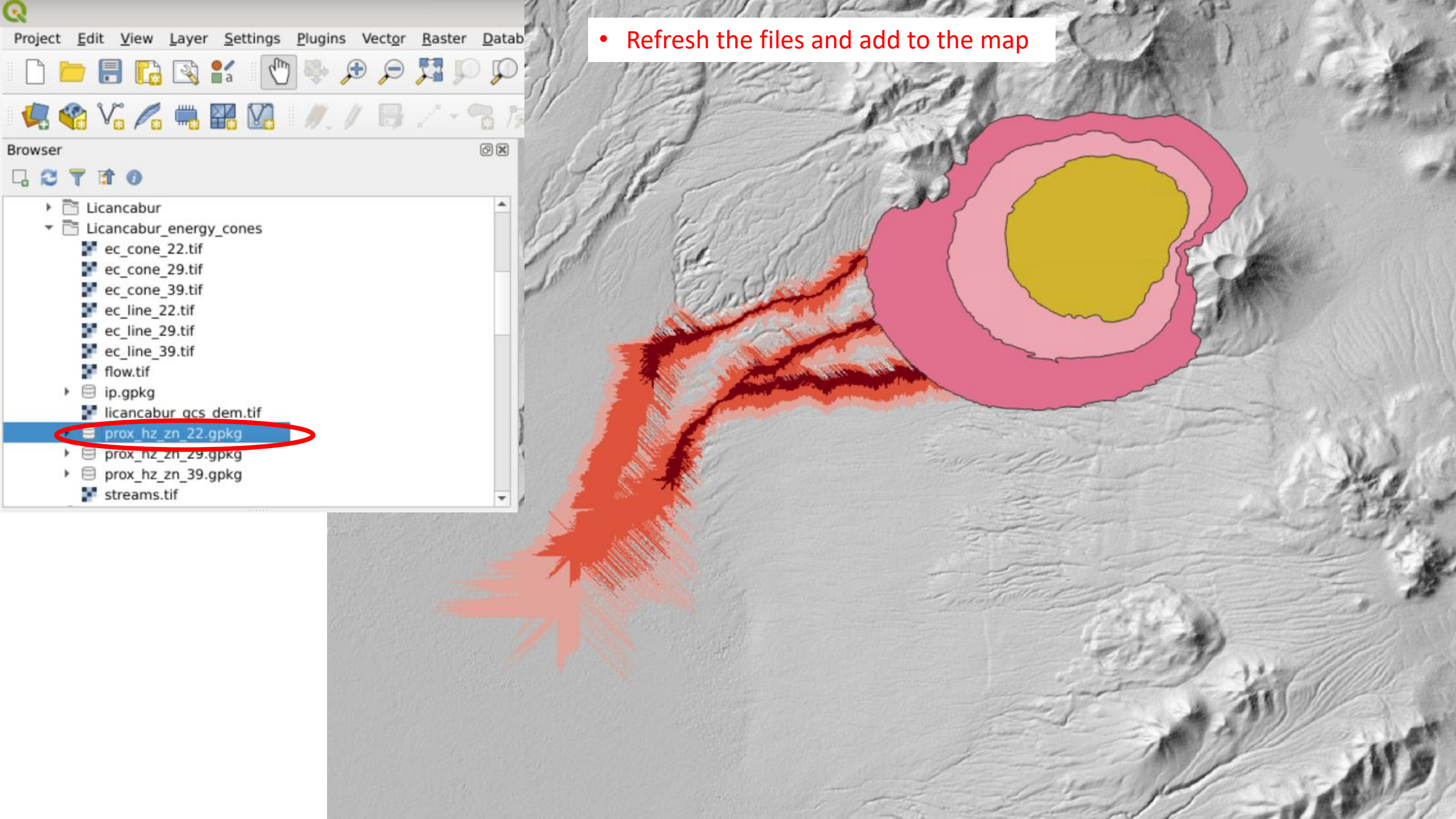


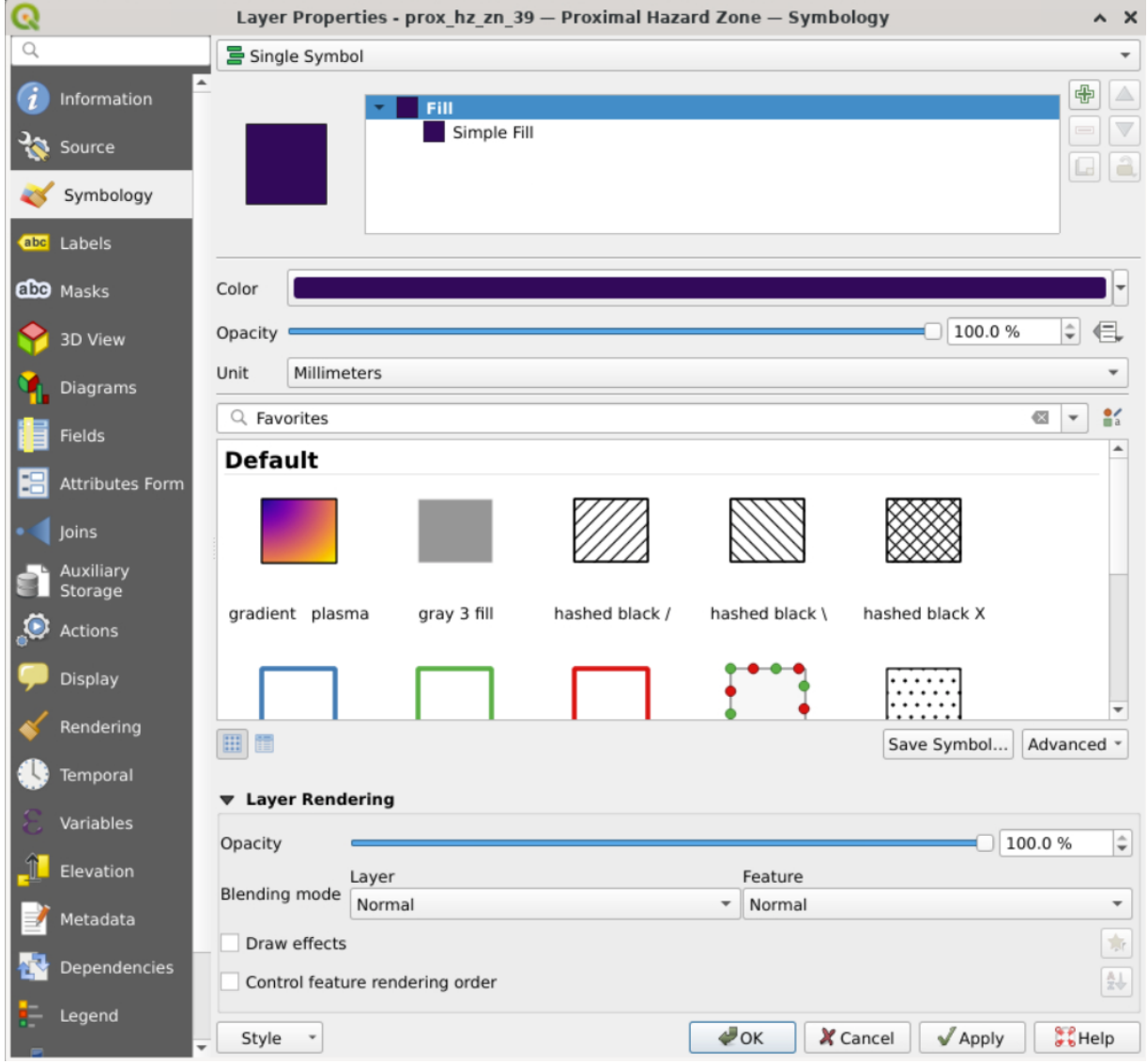
- It will warn you that the other files will be overwritten, but we don’t care, just click “overwrite”


Project Edit View Layer Settings Plugins Vector Raster Datab
Browser
Licancabur
Licancabur_energy_cones
ec_cone_29.tif
ec_cone_39.tif
ec_line_29.tif
ec_line_39.tif
flow.tif
ip.gpkg
licancabur_gcs_dem.tif
prox_hz_zn_29.gpkg
prox_hz_zn_39.gpkg
streams.tif
__init__
__main__
laharz
• Refresh the files and add to the map

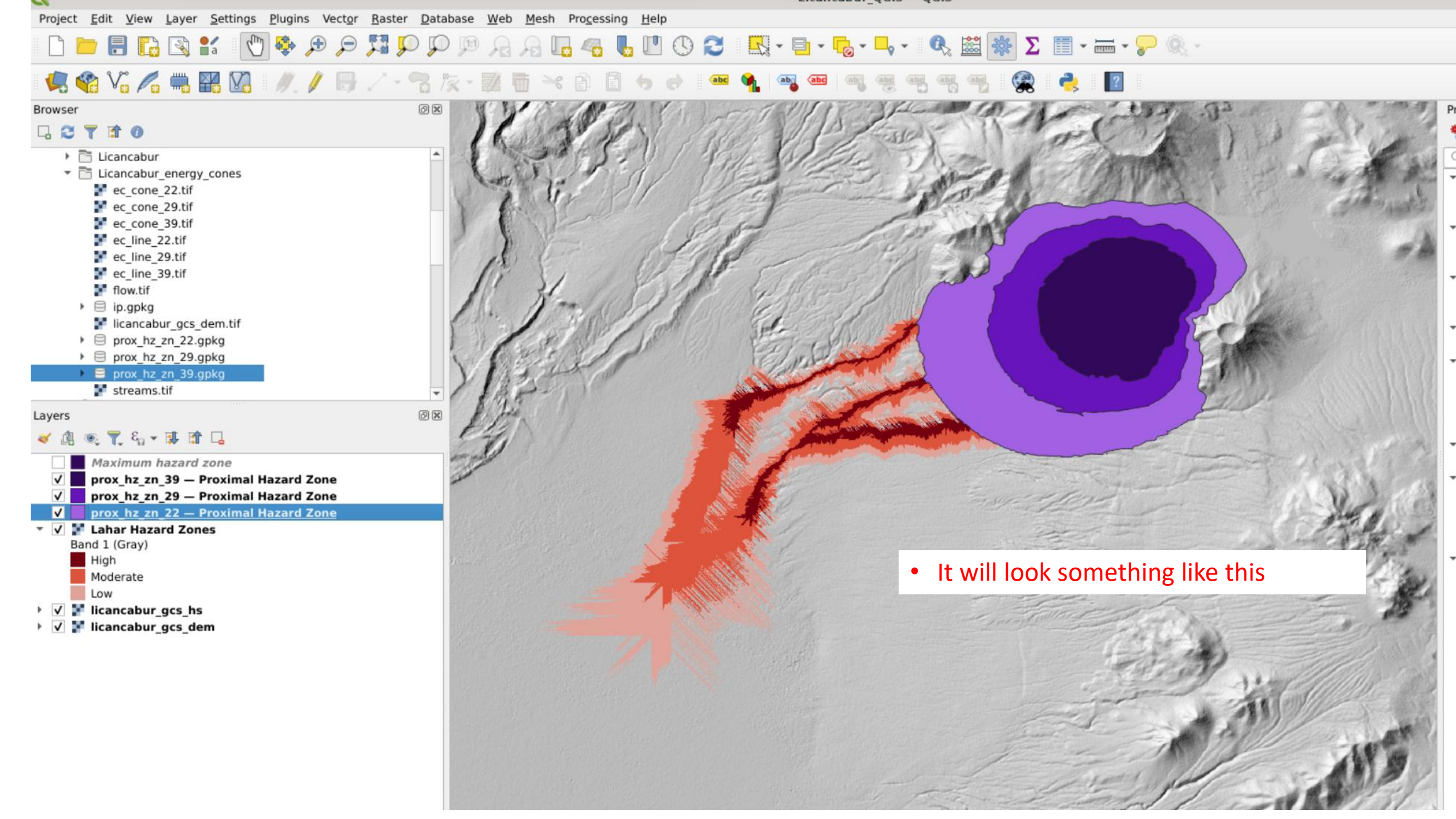


- Repeat for 0.22


Project Edit View Layer Settings Plugins Vector Raster Datab
Browser
Licancabur
Licancabur_energy_cones
ec_cone_22.tif
ec_cone_29.tif
ec_cone_39.tif
ec_line_22.tif
ec_line_29.tif
ec_line_39.tif
flow.tif
ip.gpkg
licancabur_gcs_dem.tif
prox_hz_zn_22.gpkg
prox_hz_zn_29.gpkg
prox_hz_zn_39.gpkg
streams.tif
• Refresh the files and add to the map



- Change the colors of the layers by write clicking and choosing properties, then symbology
- Lets make the zones Dark to Light purple, with the smallest zone (39) the darkest


Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Licancabur
Licancabur_energy_cones
ec_cone_22.tif
ec_cone_29.tif
ec_cone_39.tif
ec_line_22.tif
ec_line_29.tif
ec_line_39.tif
flow.tif
ip.gpkg
licancabur_gcs_dem.tif
prox_hz_zn_22.gpkg
prox_hz_zn_29.gpkg
prox_hz_zn_39.gpkg
streams.tif
Layers
Maximum hazard zone
prox_hz_zn_39 — Proximal Hazard Zone
prox_hz_zn_29 — Proximal Hazard Zone
prox_hz_zn_22 — Proximal Hazard Zone
Lahar Hazard Zones
Band 1 (Gray)
High
Moderate
Low
licancabur_gcs_hs
licancabur_gcs_dem
• It will look something like this

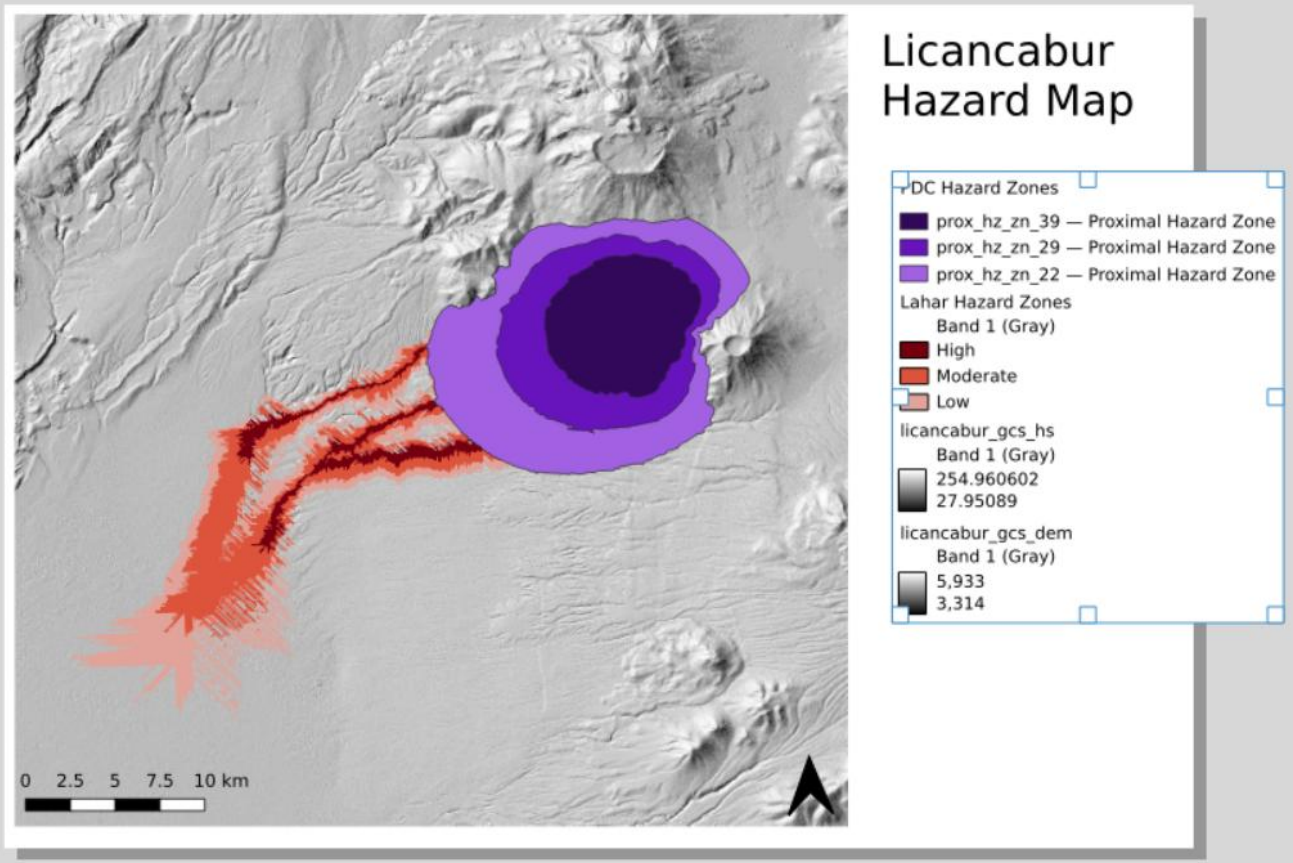
jupyter

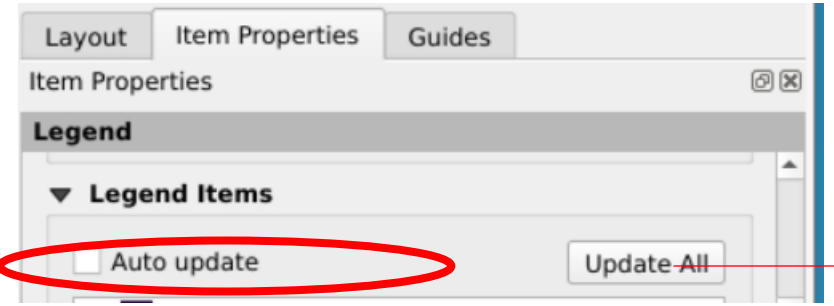
Applications | LaharZ 2.1.4 | QGIS3 | Terminal -

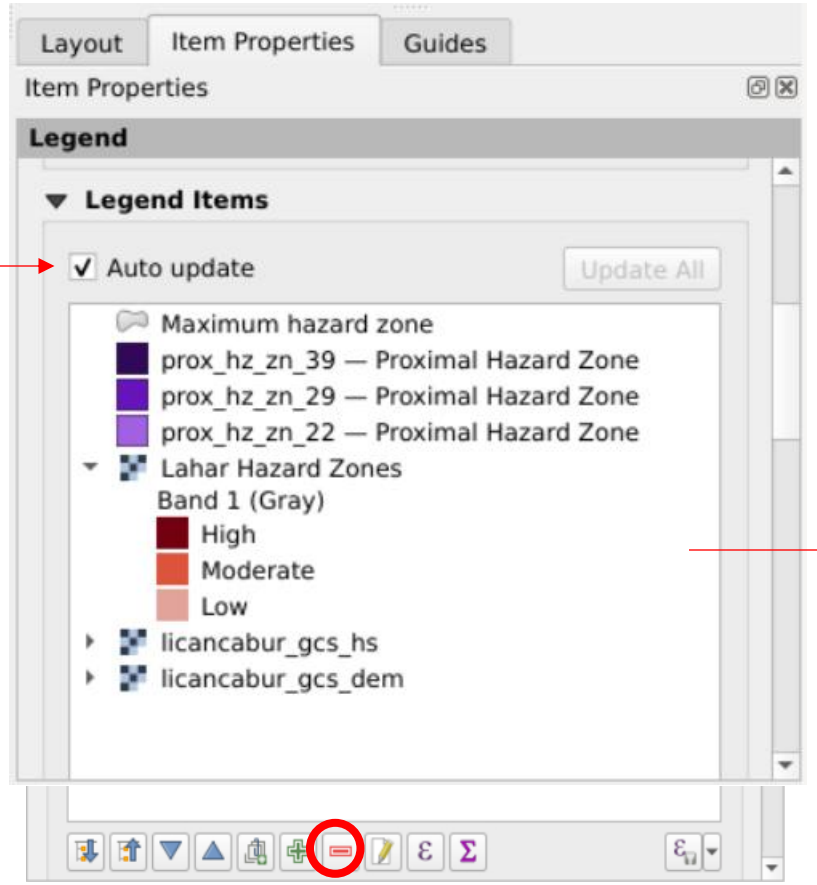
*Licancabur_hazard_map

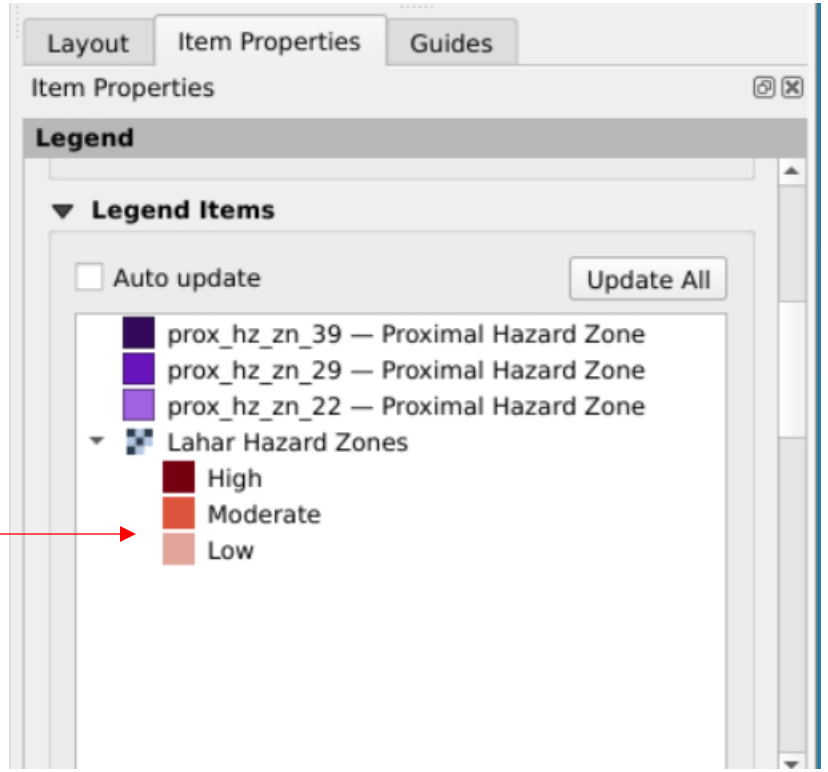
Layout Edit View Items Add Item Atlas Settings

- Let's add everything to the hazard map
- Go back to your hazard map by clicking "QGIS" at the top of the screen
- The legend will need work

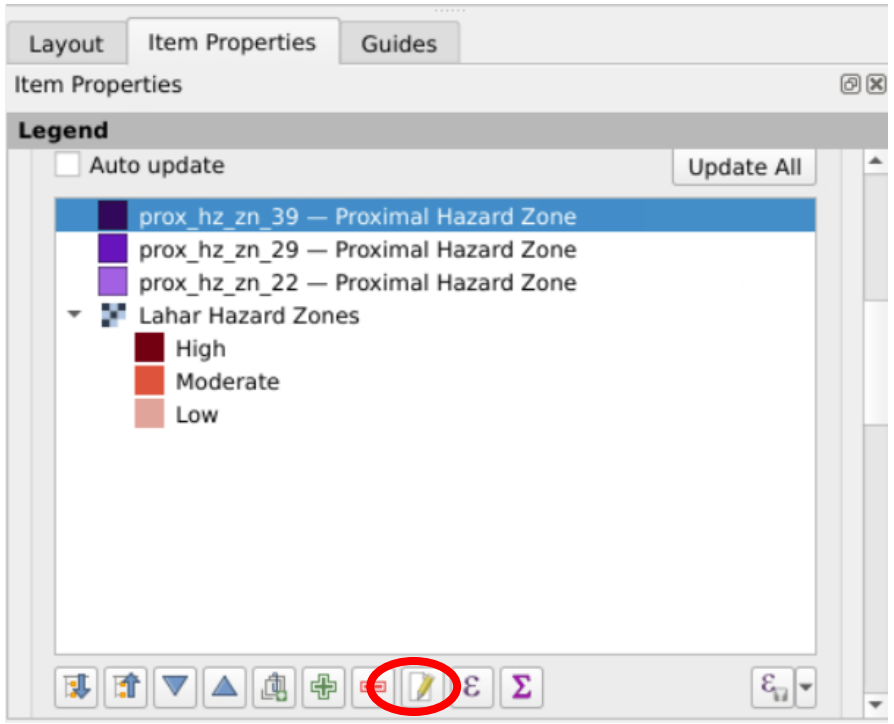


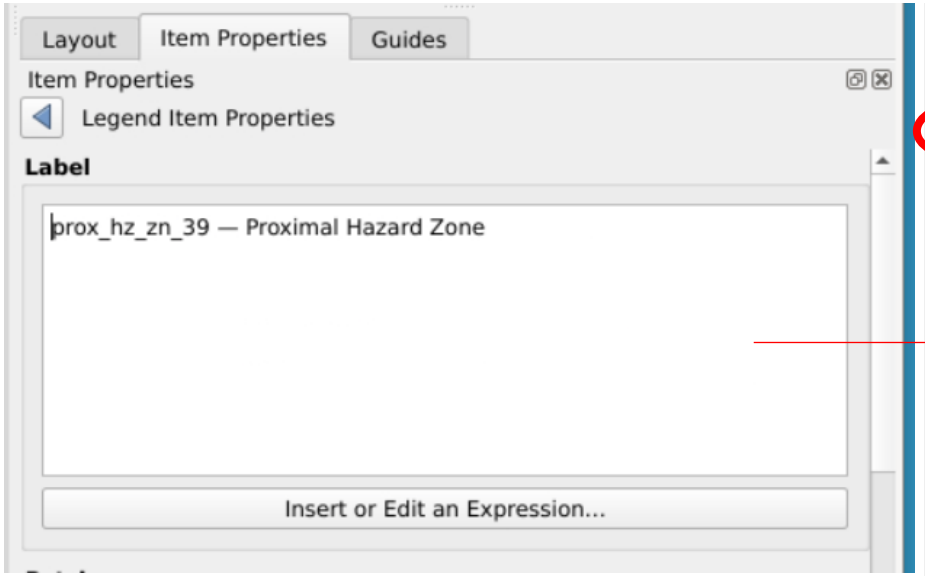


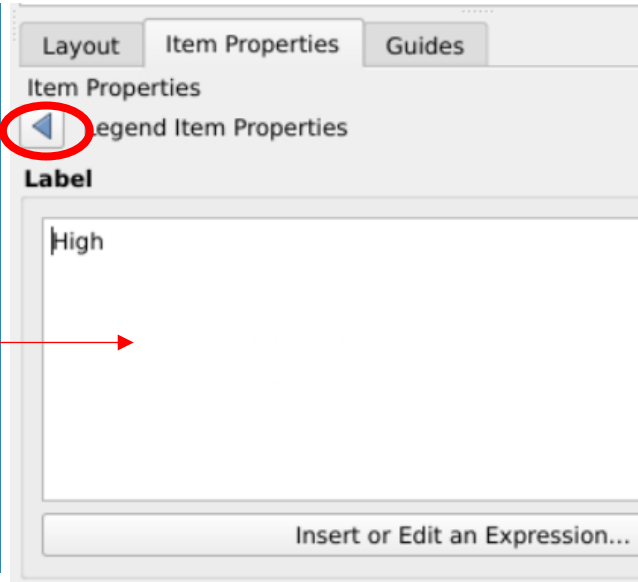




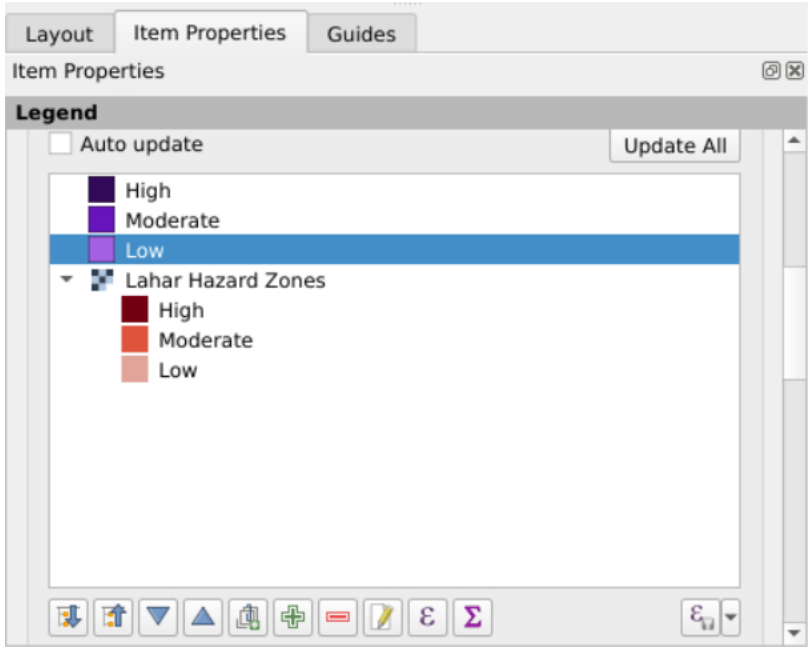
- Click on the legend and go to properties
- Turn on auto update to add the items
- Turn it back off
- Remove the maximum hazard zone, Band 1, and the DEMs and hillshade by clicking the layer and then the red minus





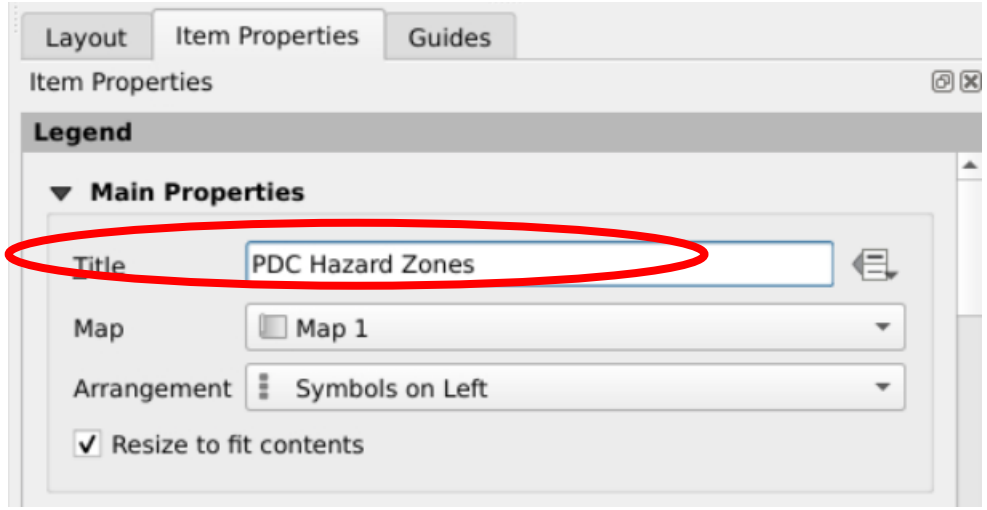


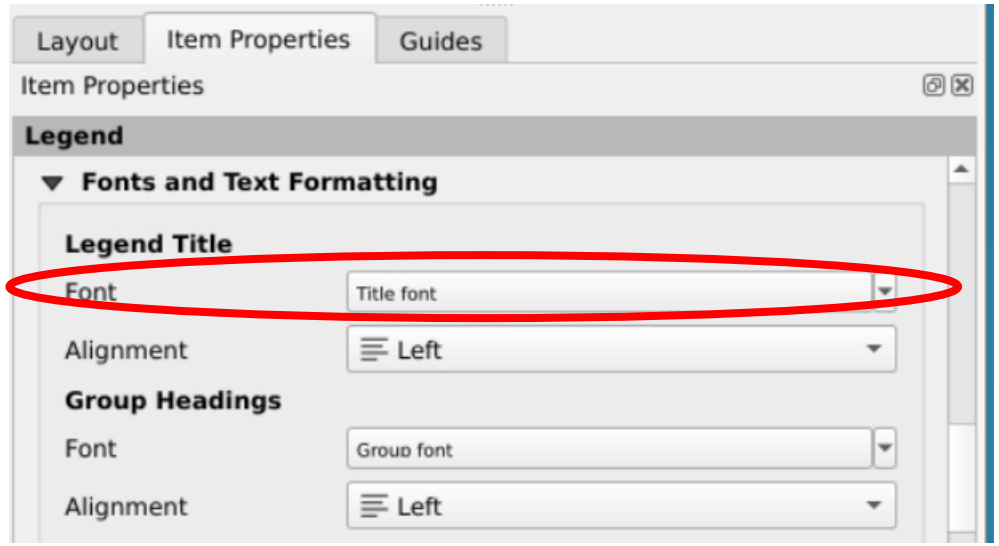
- Let's rename the PDC energy cones
- Select the layer and then the edit button
- Change the layer names to High, Moderate, and Low, with 39 = High
- Click the blue back button to continue



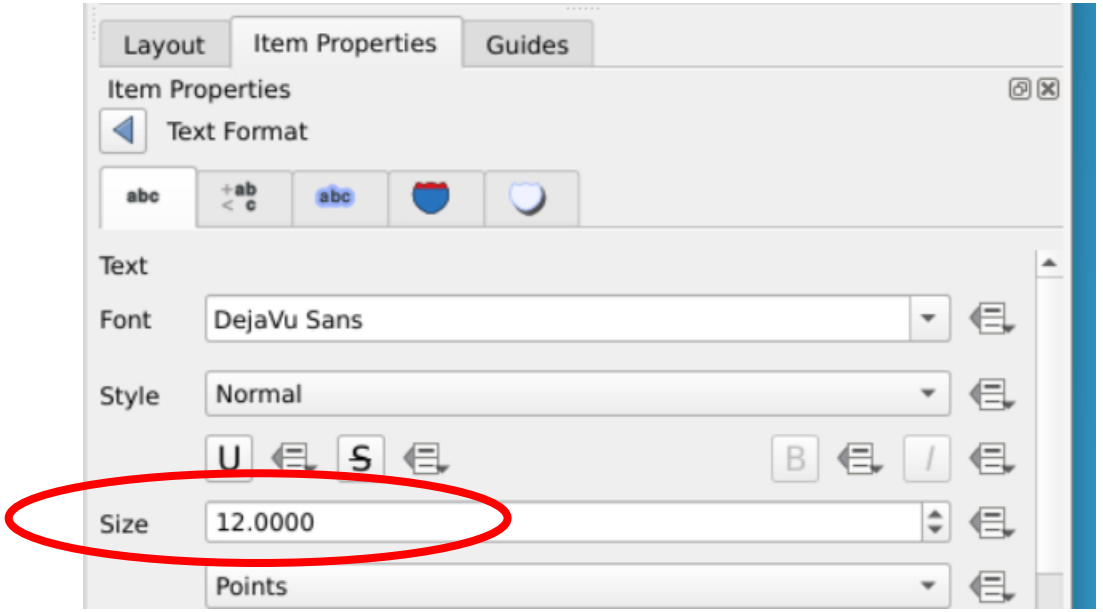


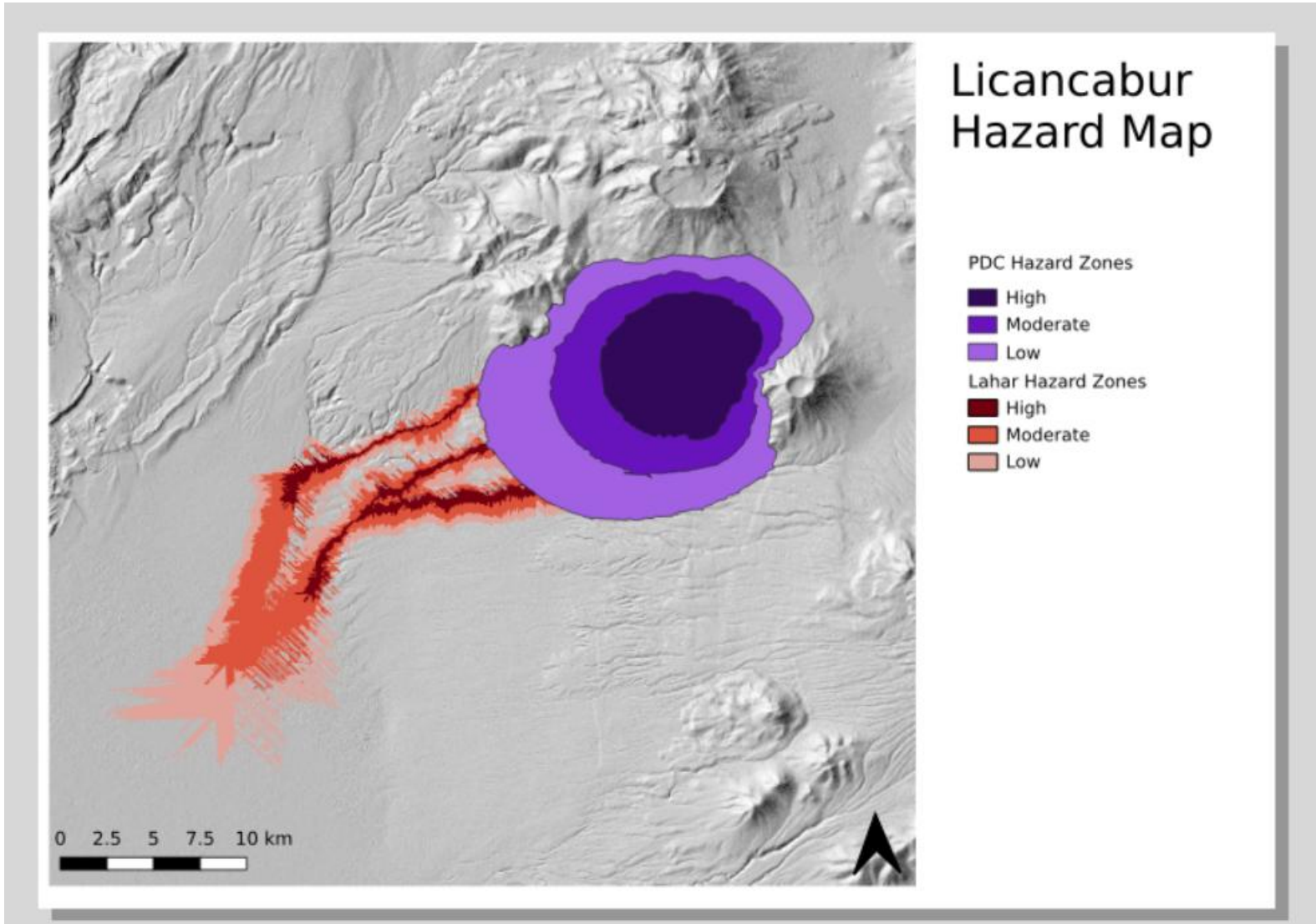
- The legend has a group name for lahars, but not for PDCs
- We could fix this in Illustrator, but let's just take a quick shortcut and change the legend name instead





- Go to the Legend Main Properties
- Change the title to “PDC Hazard Zones”
- Edit the Title Font to Size=12.000





- The final map will look something like this!